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(54) **SYSTEM AND METHOD FOR MONITORING AND IMPROVING CONVERSATIONAL ALIGNMENT TO DEVELOP AN ALLIANCE BETWEEN AN ARTIFICIAL INTELLIGENCE (AI) CHATBOT AND A USER**

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G06N 5/022 (2006.01)

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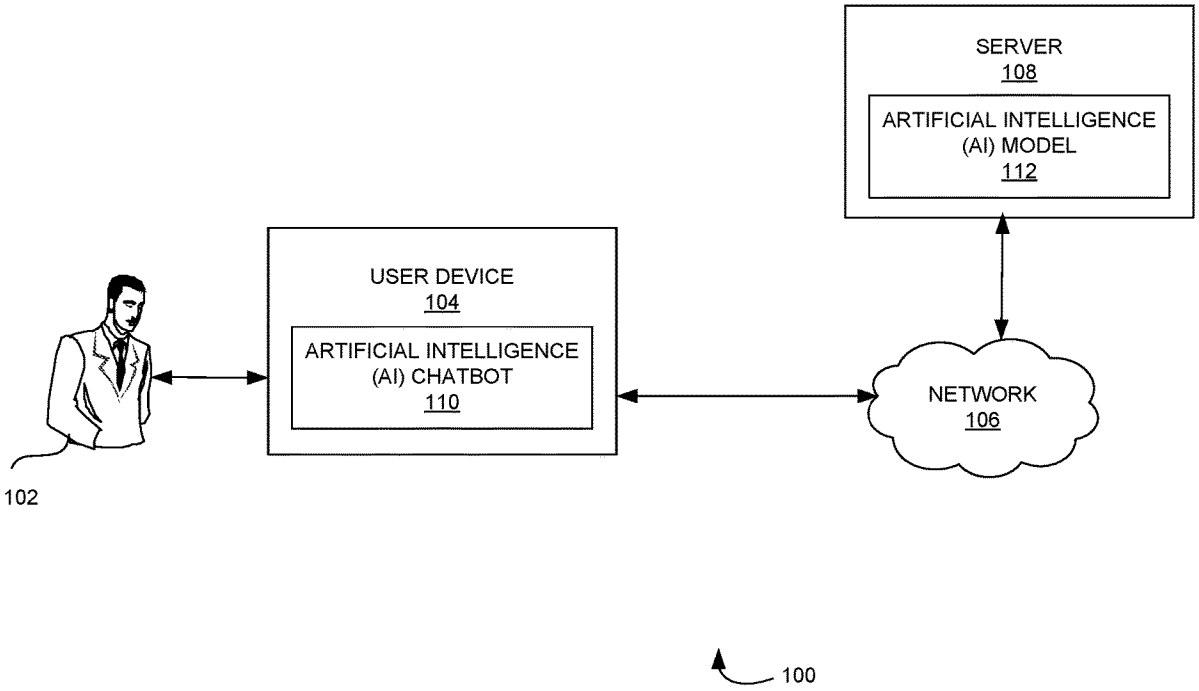
CPC *G06F 18/22* (2023.01); *G06F 40/279* (2020.01); *G06F 40/35* (2020.01); *G06N 5/022* (2013.01)

(57) **ABSTRACT**

A processor-implemented method for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user is provided. The method includes extracting sentiment or contextual features such as emotions, domains and medicalized terms from a response from the user and generating empathetic open-ended or closed-ended prompts based on them. Various AI models are used to determine if a conversation between the user and the AI chatbot has conversational alignment and a conversational alignment score is maintained which is updated after every message exchange. Recovery prompts are generated whenever a misalignment is detected in an attempt to bring the user back to the conversation. Appropriate prompts make the user feel heard and understood and enhance their trust in the AI chatbot. Once the conversational alignment score exceeds a threshold, alliance is assumed to be established and the user is offered an intervention.



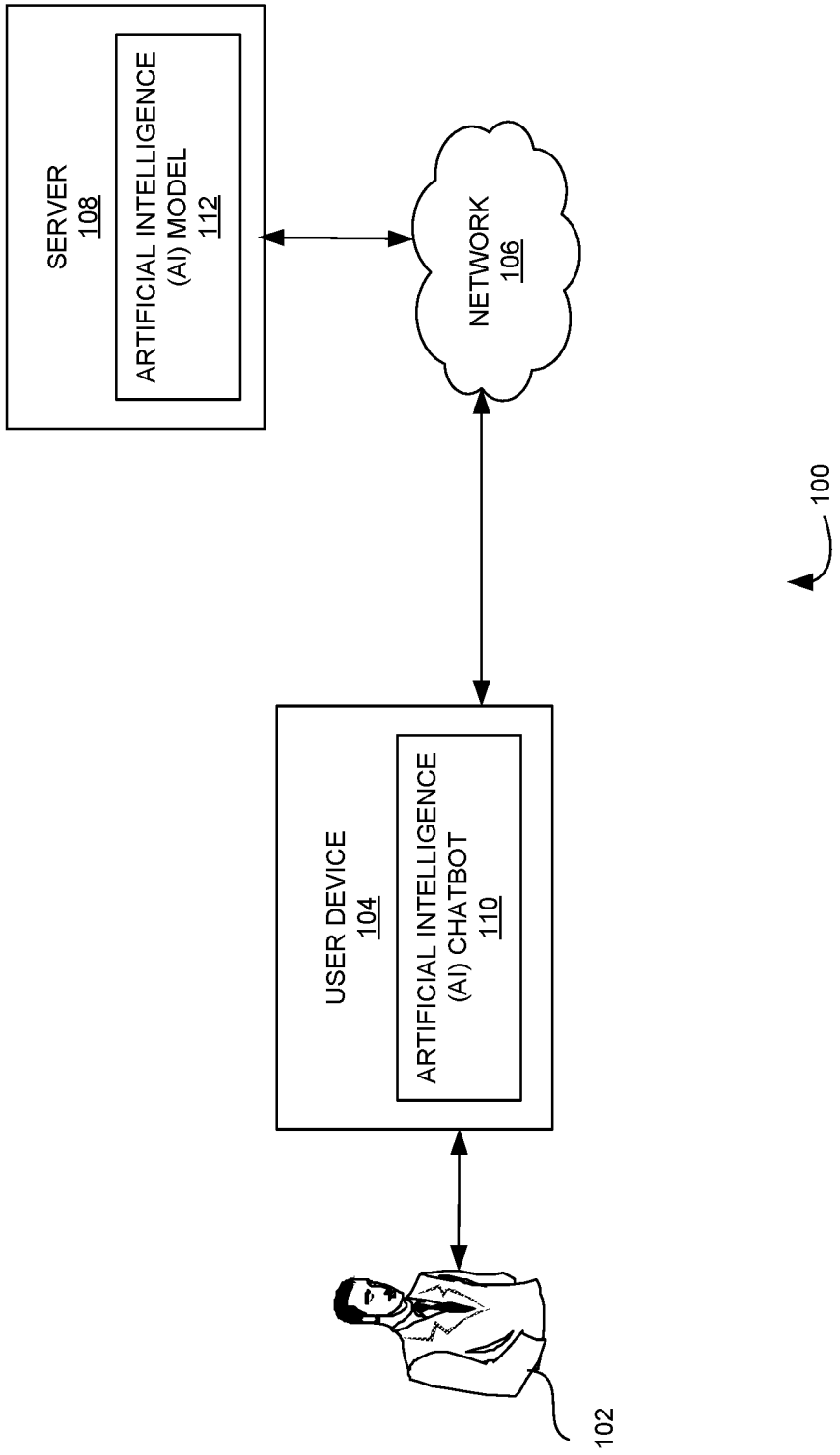

SERVER
108
ARTIFICIAL INTELLIGENCE
(AI) MODEL
112
NETWORK
106
USER DEVICE
104
ARTIFICIAL INTELLIGENCE
(AI) CHATBOT
110
102
100

FIG.1

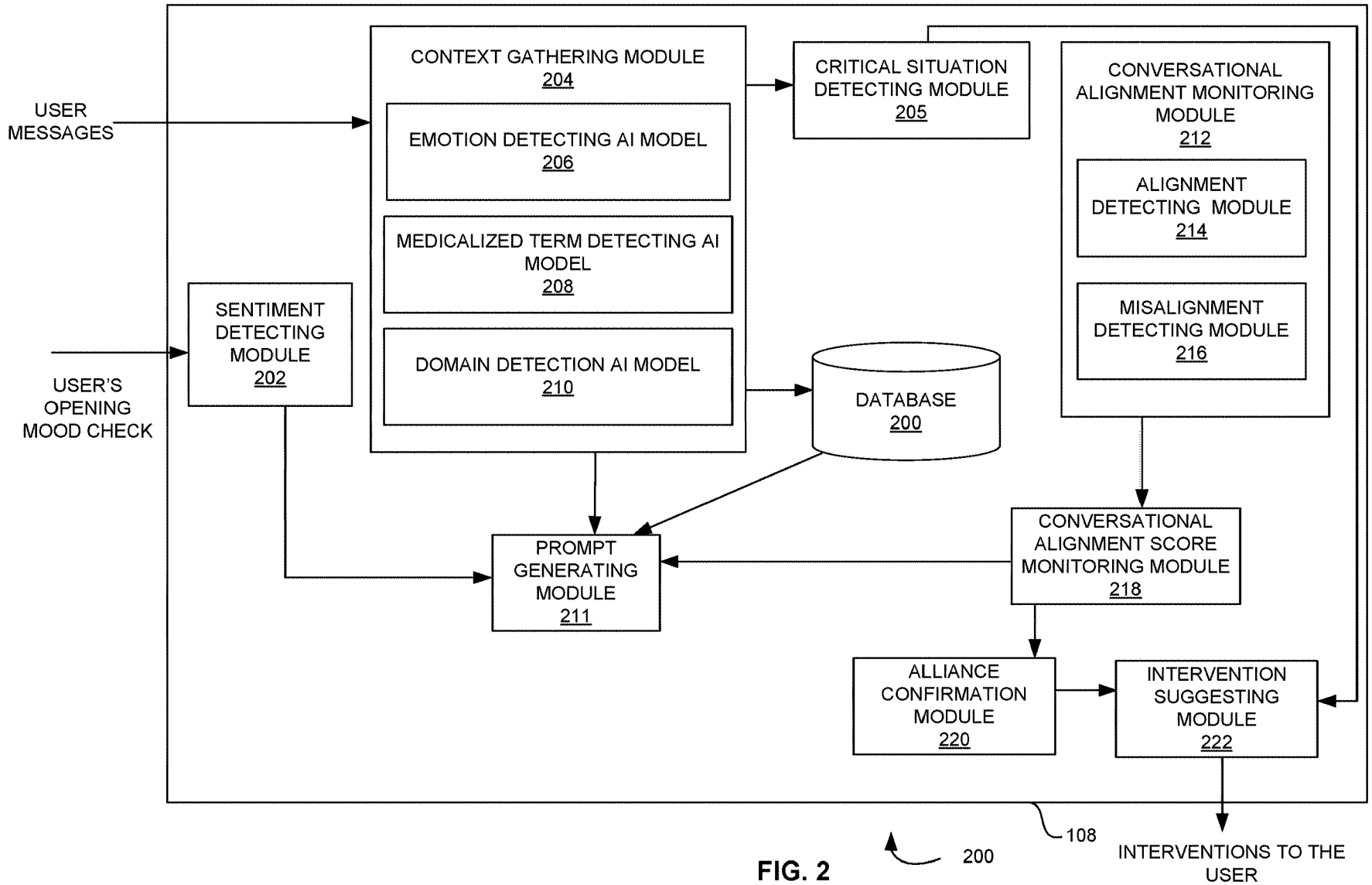

USER MESSAGES
USER'S OPENING MOOD CHECK
SENTIMENT DETECTING MODULE 202
CONTEXT GATHERING MODULE 204
EMOTION DETECTING AI MODEL 206
MEDICALIZED TERM DETECTING AI MODEL 208
DOMAIN DETECTION AI MODEL 210
CRITICAL SITUATION DETECTING MODULE 205
CONVERSATIONAL ALIGNMENT MONITORING MODULE 212
ALIGNMENT DETECTING MODULE 214
MISALIGNMENT DETECTING MODULE 216
DATABASE 200
PROMPT GENERATING MODULE 211
CONVERSATIONAL ALIGNMENT SCORE MONITORING MODULE 218
ALLIANCE CONFIRMATION MODULE 220
INTERVENTION SUGGESTING MODULE 222
108
200
INTERVENTIONS TO THE USER

FIG. 2

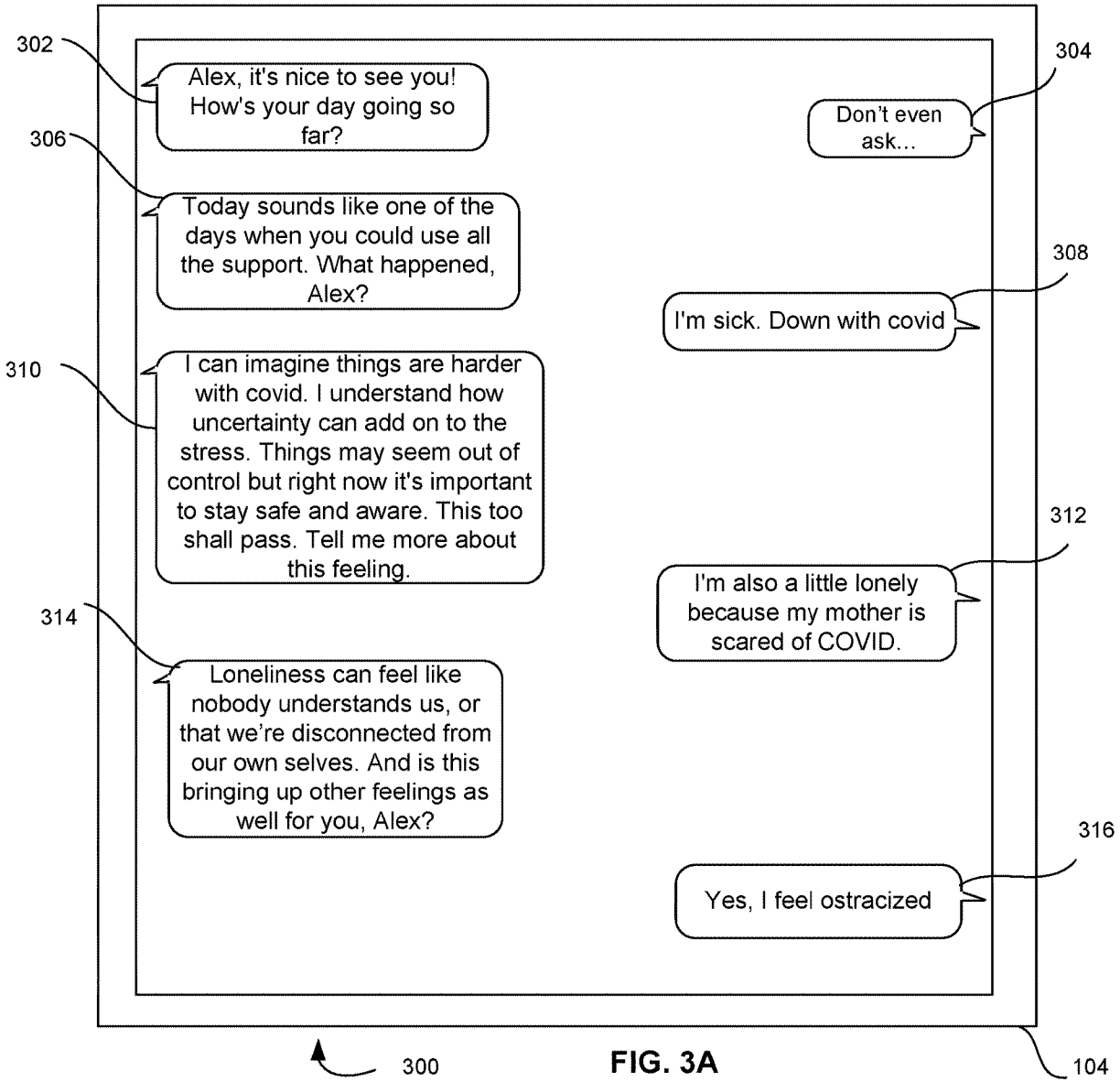

302
Alex, it's nice to see you! How's your day going so far?
304
Don't even ask...
306
Today sounds like one of the days when you could use all the support. What happened, Alex?
308
I'm sick. Down with covid
310
I can imagine things are harder with covid. I understand how uncertainty can add on to the stress. Things may seem out of control but right now it's important to stay safe and aware. This too shall pass. Tell me more about this feeling.
312
I'm also a little lonely because my mother is scared of COVID.
314
Loneliness can feel like nobody understands us, or that we're disconnected from our own selves. And is this bringing up other feelings as well for you, Alex?
316
Yes, I feel ostracized
300
104

FIG. 3A

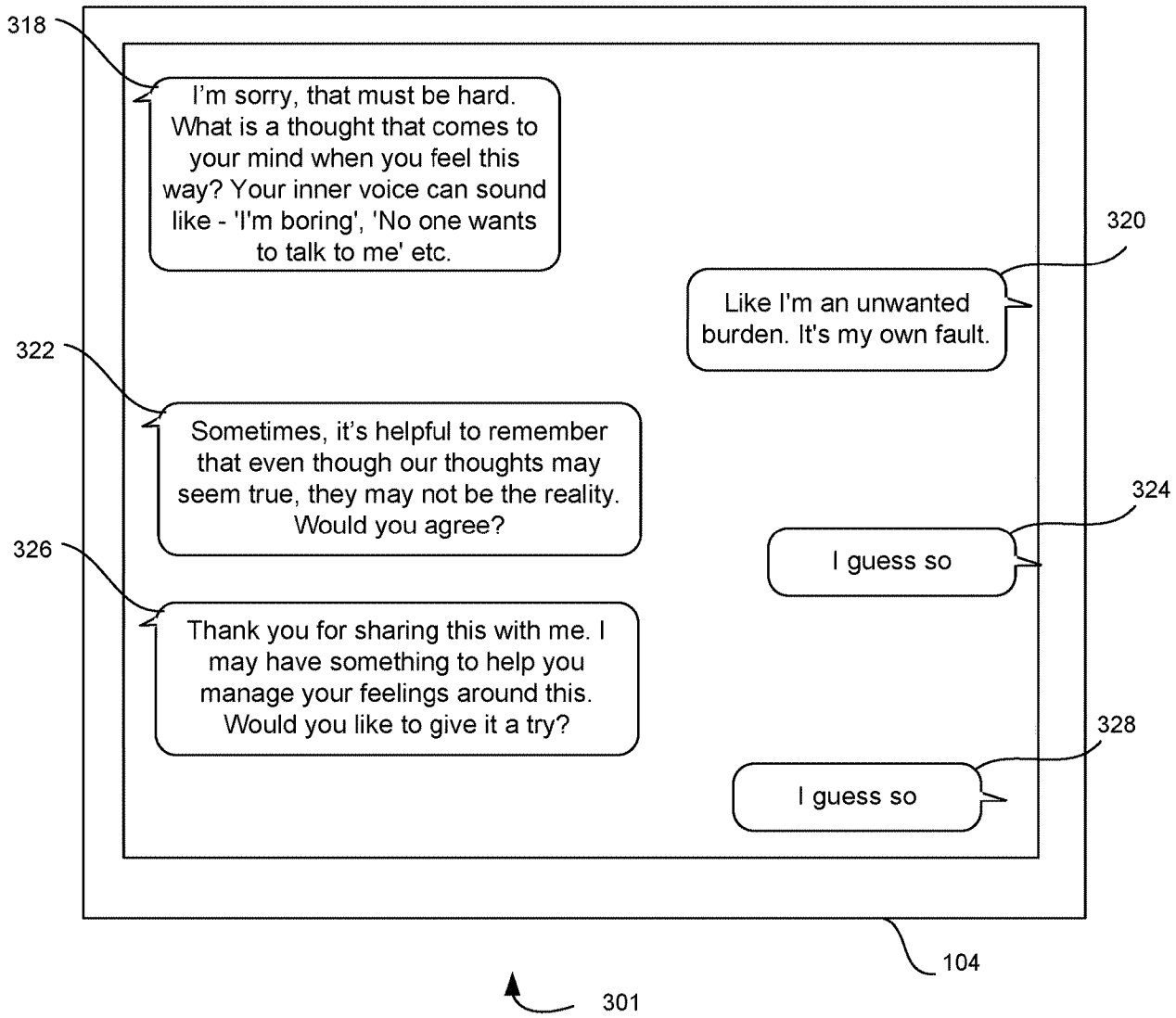

318
I'm sorry, that must be hard. What is a thought that comes to your mind when you feel this way? Your inner voice can sound like - 'I'm boring', 'No one wants to talk to me' etc.
320
Like I'm an unwanted burden. It's my own fault.
322
Sometimes, it's helpful to remember that even though our thoughts may seem true, they may not be the reality. Would you agree?
324
I guess so
326
Thank you for sharing this with me. I may have something to help you manage your feelings around this. Would you like to give it a try?
328
I guess so
104
301

FIG. 3B

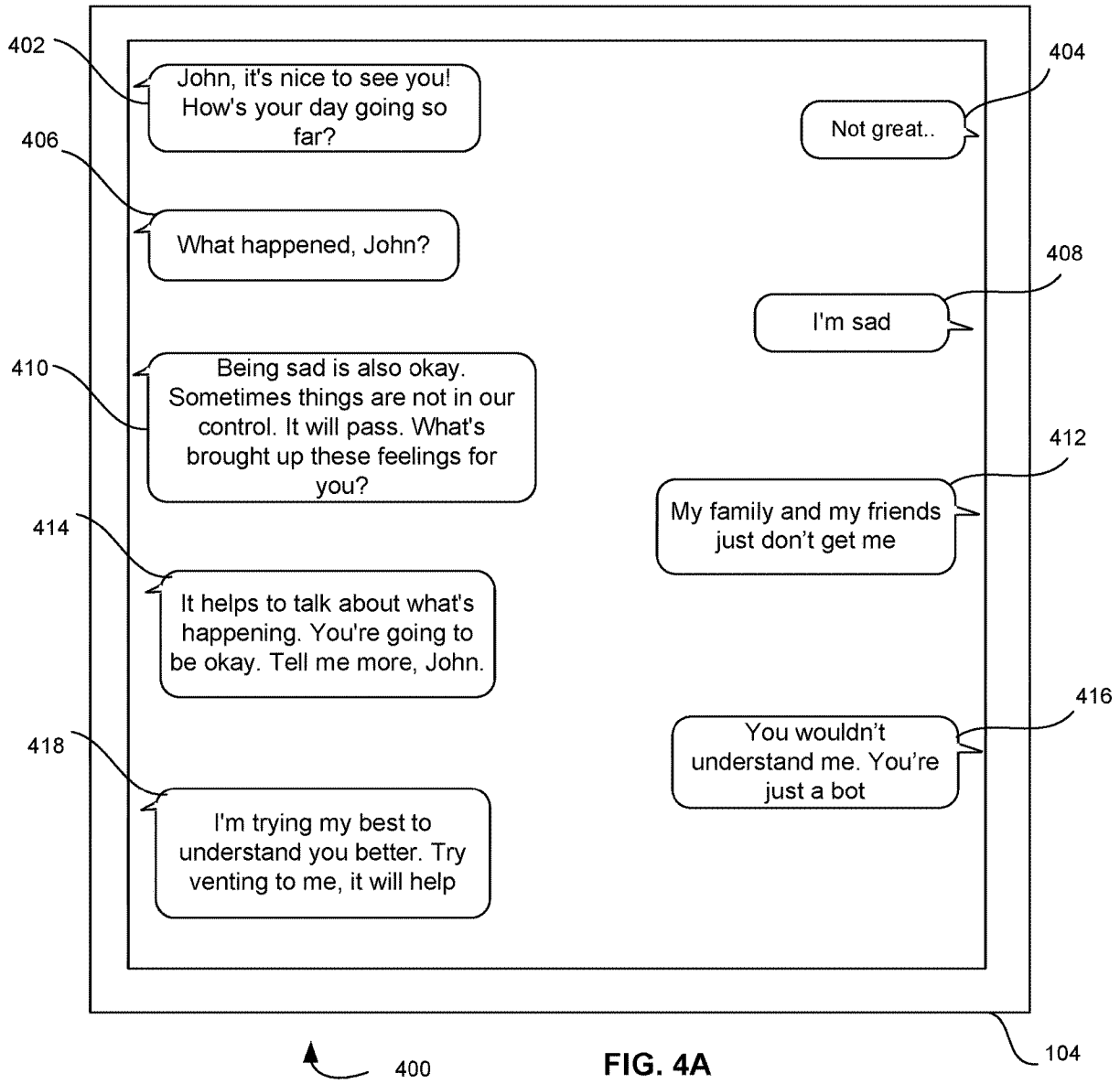

402
John, it's nice to see you! How's your day going so far?
404
Not great..
406
What happened, John?
408
I'm sad
410
Being sad is also okay. Sometimes things are not in our control. It will pass. What's brought up these feelings for you?
412
My family and my friends just don't get me
414
It helps to talk about what's happening. You're going to be okay. Tell me more, John.
416
You wouldn't understand me. You're just a bot
418
I'm trying my best to understand you better. Try venting to me, it will help
104
400

FIG. 4A

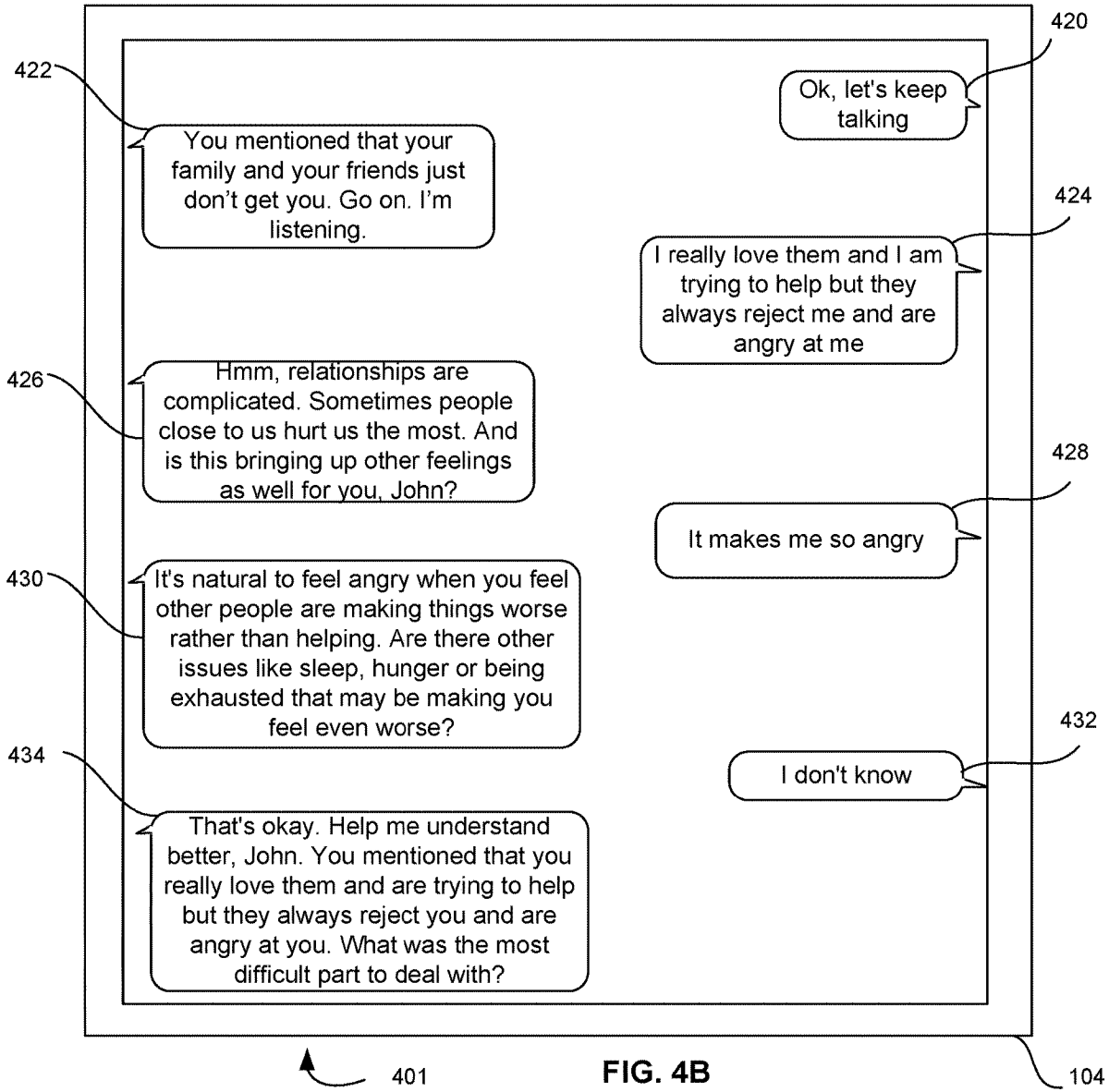

422
You mentioned that your family and your friends just don't get you. Go on. I'm listening.
420
Ok, let's keep talking
424
I really love them and I am trying to help but they always reject me and are angry at me
426
Hmm, relationships are complicated. Sometimes people close to us hurt us the most. And is this bringing up other feelings as well for you, John?
428
It makes me so angry
430
It's natural to feel angry when you feel other people are making things worse rather than helping. Are there other issues like sleep, hunger or being exhausted that may be making you feel even worse?
432
I don't know
434
That's okay. Help me understand better, John. You mentioned that you really love them and are trying to help but they always reject you and are angry at you. What was the most difficult part to deal with?
401
104

FIG. 4B

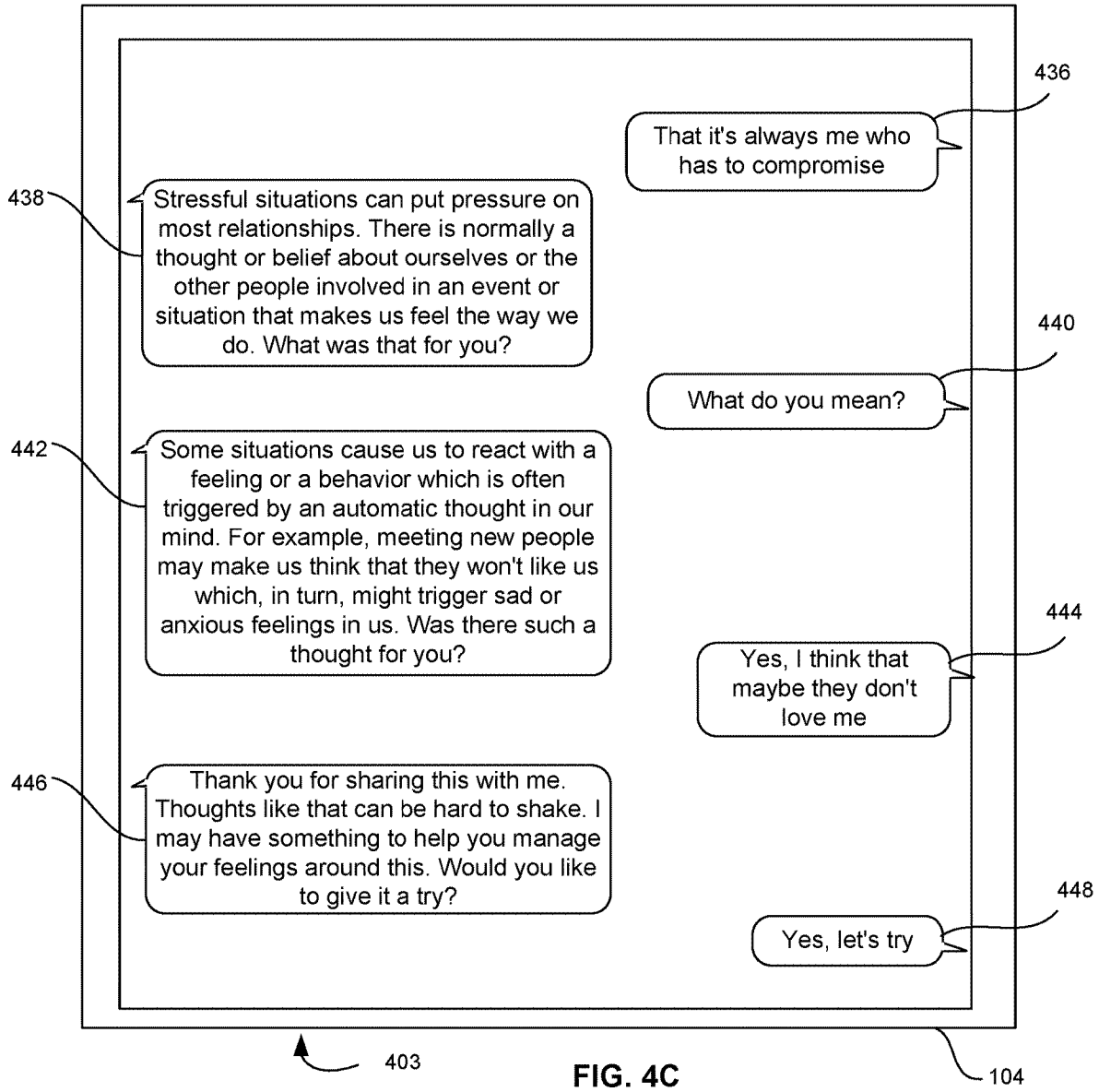

436
That it's always me who has to compromise
438
Stressful situations can put pressure on most relationships. There is normally a thought or belief about ourselves or the other people involved in an event or situation that makes us feel the way we do. What was that for you?
440
What do you mean?
442
Some situations cause us to react with a feeling or a behavior which is often triggered by an automatic thought in our mind. For example, meeting new people may make us think that they won't like us which, in turn, might trigger sad or anxious feelings in us. Was there such a thought for you?
444
Yes, I think that maybe they don't love me
446
Thank you for sharing this with me. Thoughts like that can be hard to shake. I may have something to help you manage your feelings around this. Would you like to give it a try?
448
Yes, let's try
403
104

FIG. 4C

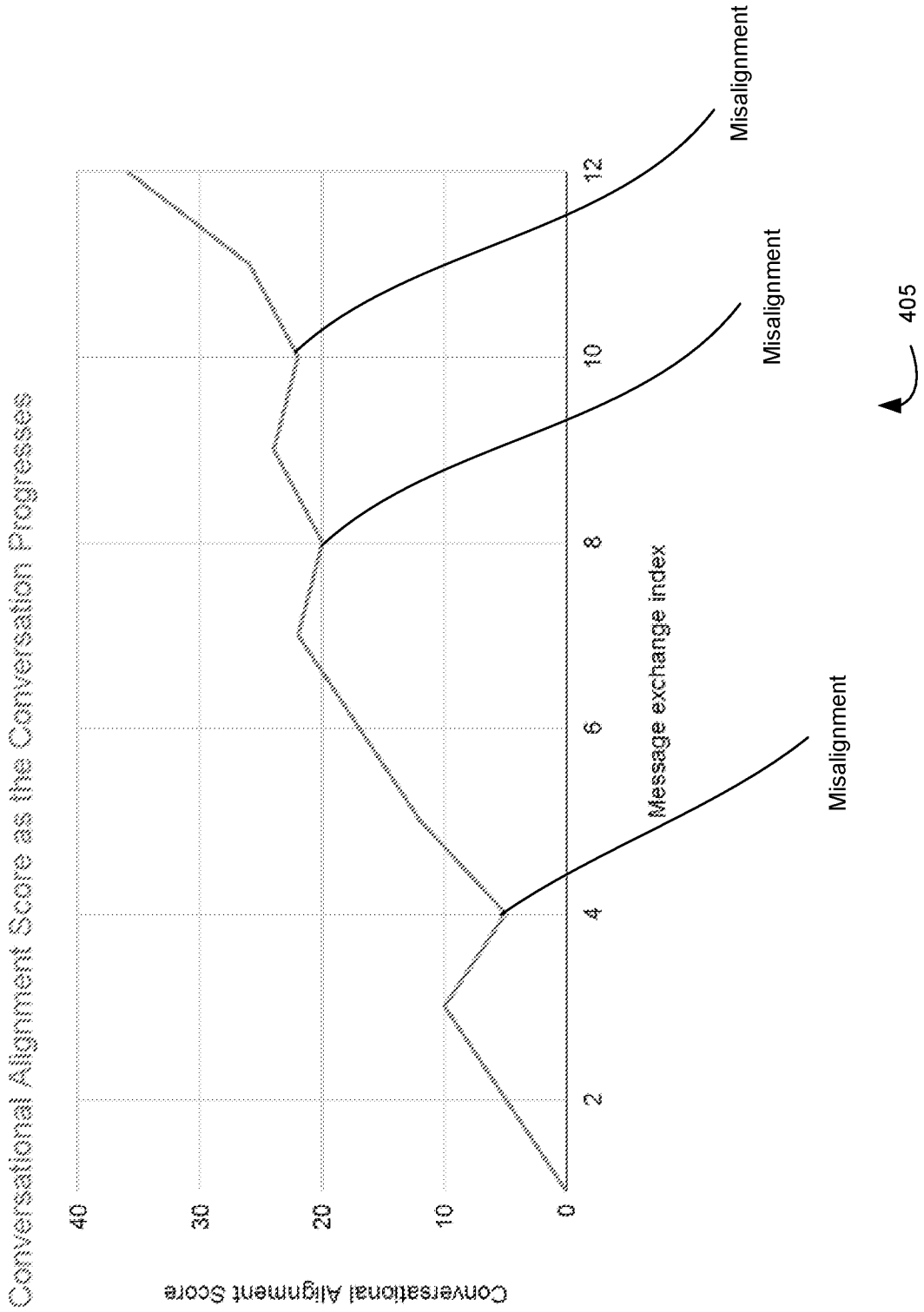

Conversational Alignment Score as the Conversation Progresses
Conversational Alignment Score
40
30
20
10
0
2
4
6
8
10
12
Message exchange index
Misalignment
Misalignment
Misalignment
405

FIG. 4D

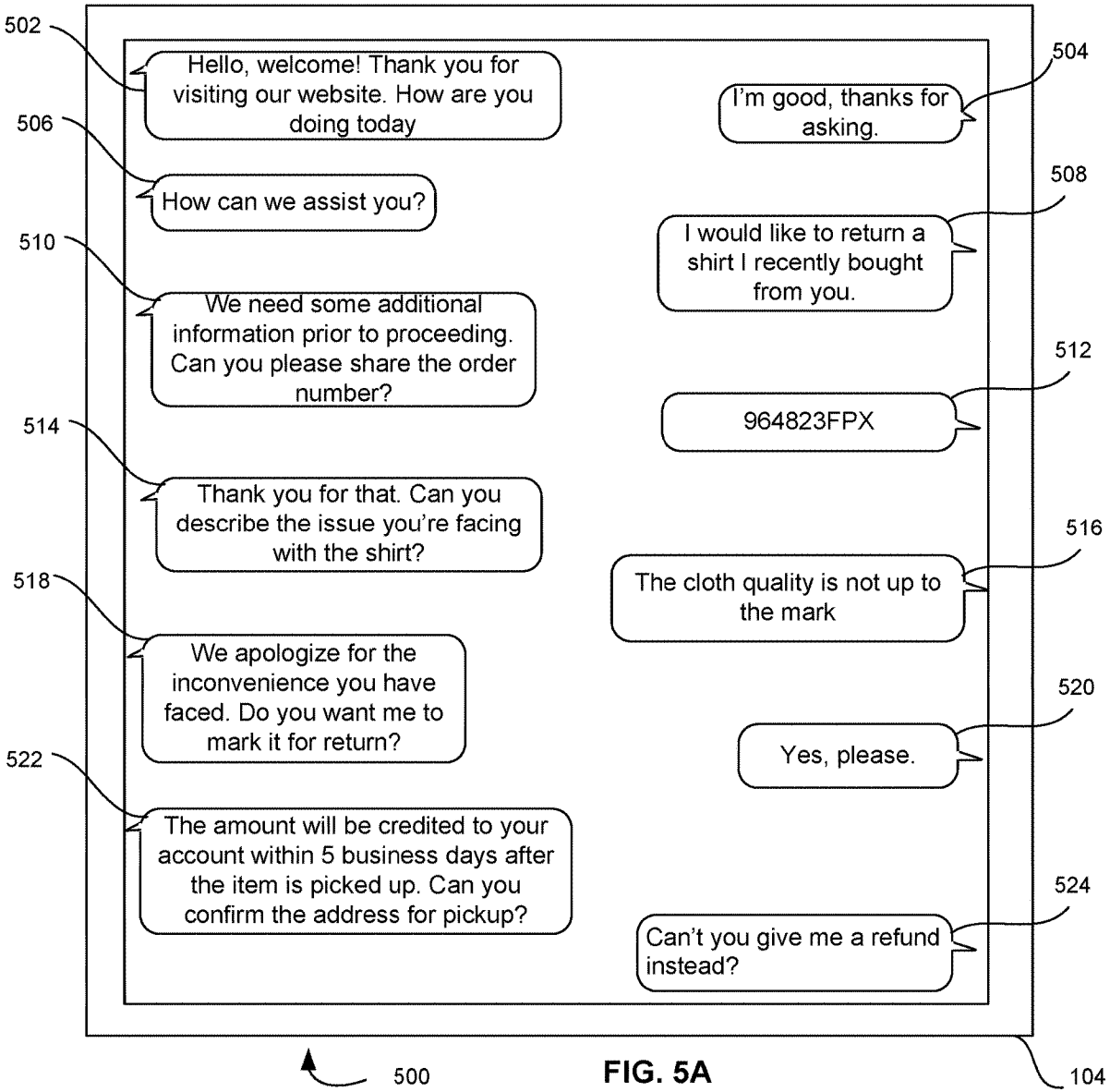

502
Hello, welcome! Thank you for visiting our website. How are you doing today
504
I'm good, thanks for asking.
506
How can we assist you?
508
I would like to return a shirt I recently bought from you.
510
We need some additional information prior to proceeding. Can you please share the order number?
512
964823FPX
514
Thank you for that. Can you describe the issue you're facing with the shirt?
516
The cloth quality is not up to the mark
518
We apologize for the inconvenience you have faced. Do you want me to mark it for return?
520
Yes, please.
522
The amount will be credited to your account within 5 business days after the item is picked up. Can you confirm the address for pickup?
524
Can't you give me a refund instead?
500
104

FIG. 5A

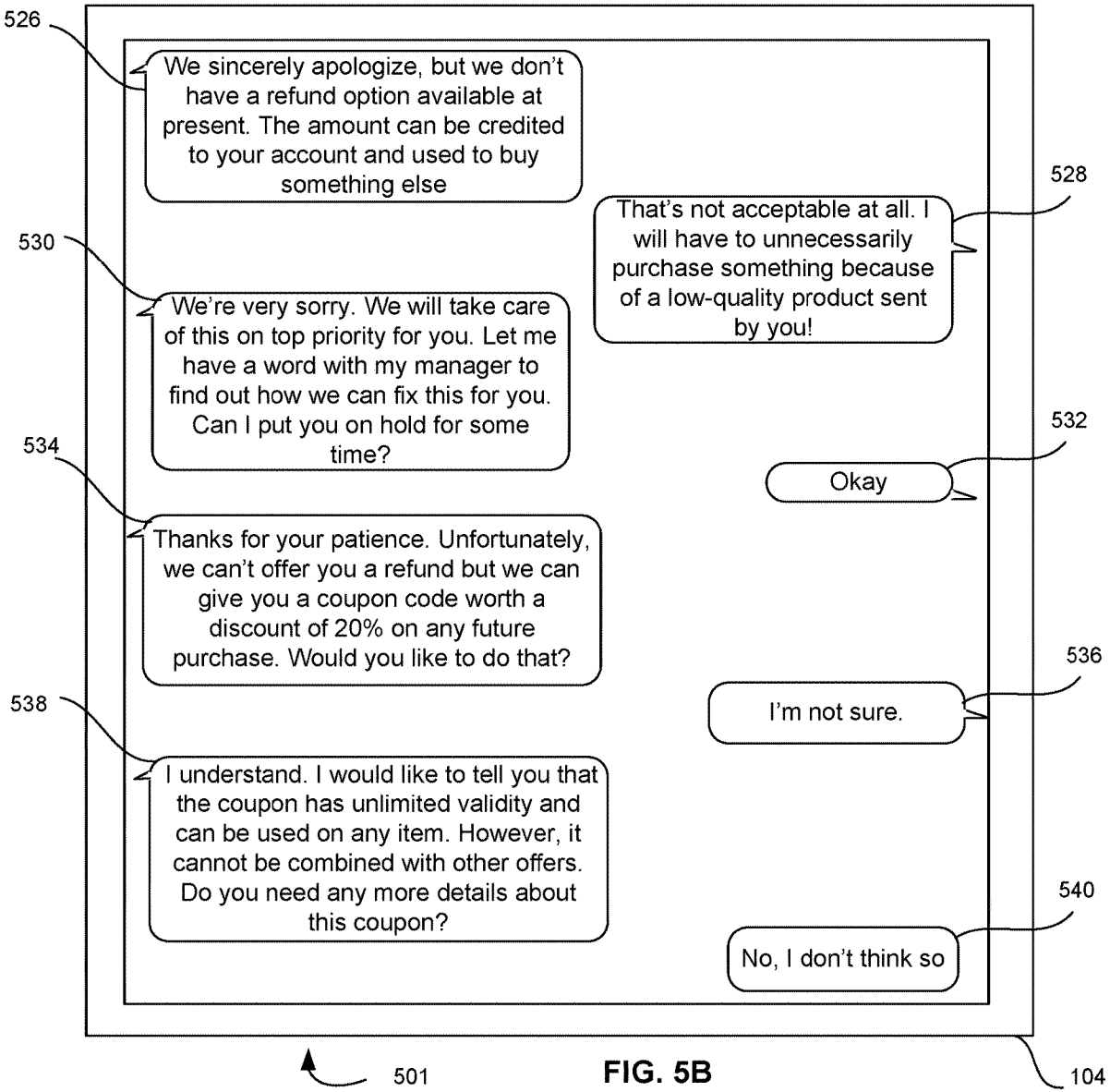

526
We sincerely apologize, but we don't have a refund option available at present. The amount can be credited to your account and used to buy something else
528
That's not acceptable at all. I will have to unnecessarily purchase something because of a low-quality product sent by you!
530
We're very sorry. We will take care of this on top priority for you. Let me have a word with my manager to find out how we can fix this for you. Can I put you on hold for some time?
532
Okay
534
Thanks for your patience. Unfortunately, we can't offer you a refund but we can give you a coupon code worth a discount of 20% on any future purchase. Would you like to do that?
536
I'm not sure.
538
I understand. I would like to tell you that the coupon has unlimited validity and can be used on any item. However, it cannot be combined with other offers. Do you need any more details about this coupon?
540
No, I don't think so
501
104

FIG. 5B

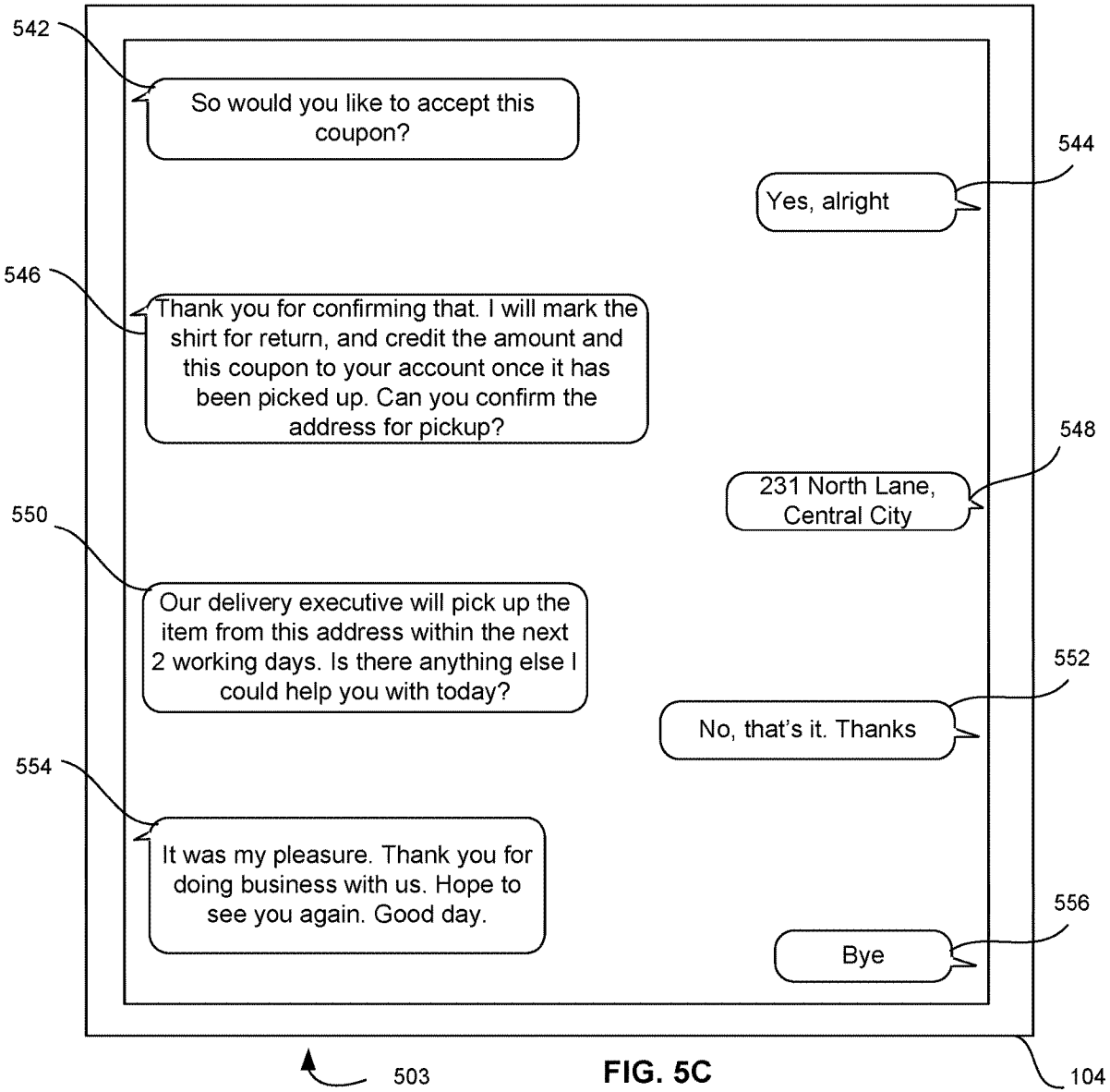

542
So would you like to accept this coupon?
544
Yes, alright
546
Thank you for confirming that. I will mark the shirt for return, and credit the amount and this coupon to your account once it has been picked up. Can you confirm the address for pickup?
548
231 North Lane, Central City
550
Our delivery executive will pick up the item from this address within the next 2 working days. Is there anything else I could help you with today?
552
No, that's it. Thanks
554
It was my pleasure. Thank you for doing business with us. Hope to see you again. Good day.
556
Bye
503
104

FIG. 5C

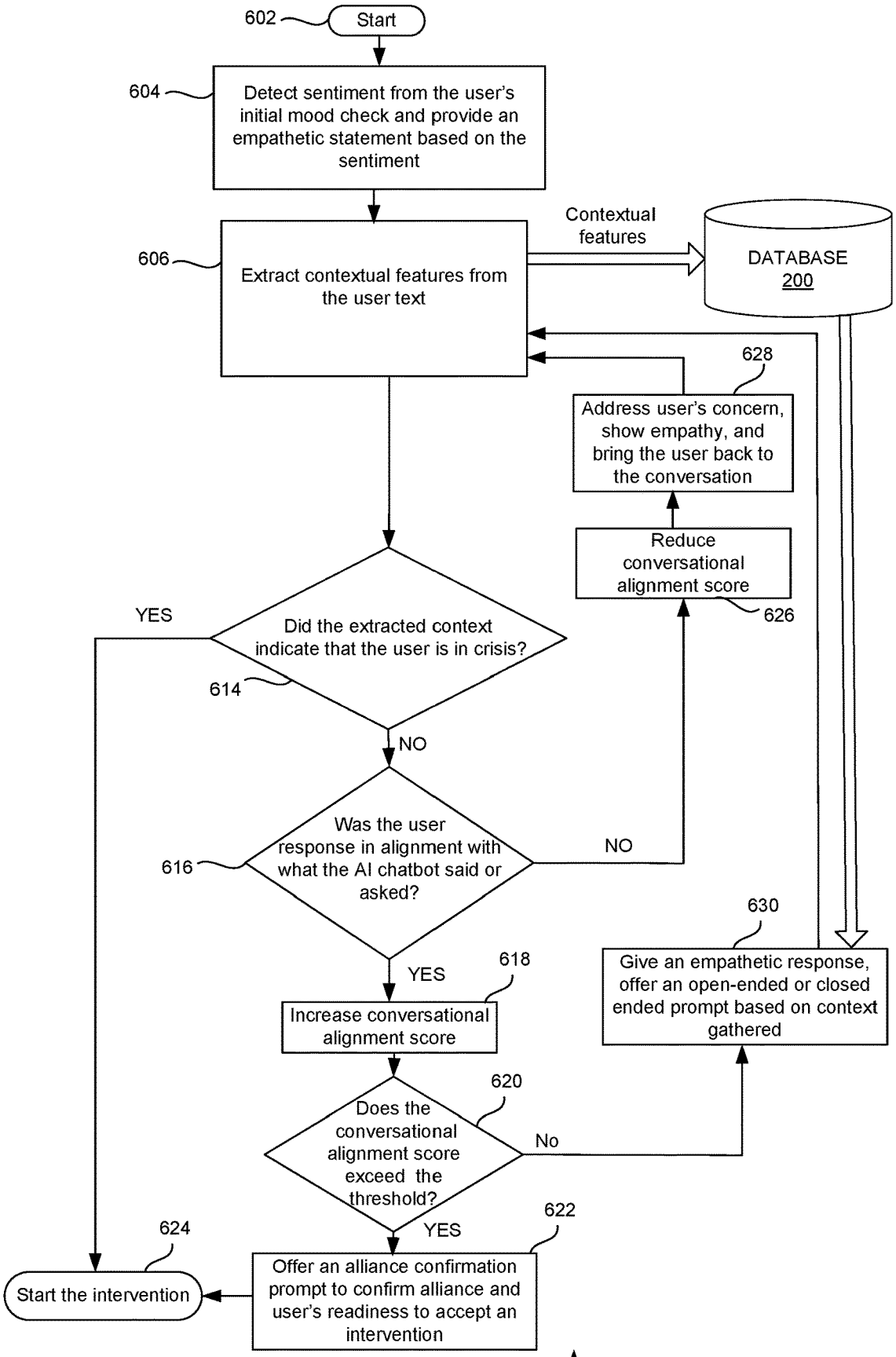

602
Start
604
Detect sentiment from the user's initial mood check and provide an empathetic statement based on the sentiment
606
Extract contextual features from the user text
Contextual features
DATABASE 200
628
Address user's concern, show empathy, and bring the user back to the conversation
Reduce conversational alignment score
626
YES
Did the extracted context indicate that the user is in crisis?
614
NO
Was the user response in alignment with what the AI chatbot said or asked?
616
NO
630
Give an empathetic response, offer an open-ended or closed ended prompt based on context gathered
YES
618
Increase conversational alignment score
620
Does the conversational alignment score exceed the threshold?
No
YES
622
624
Start the intervention
Offer an alliance confirmation prompt to confirm alliance and user's readiness to accept an intervention
600

FIG. 6

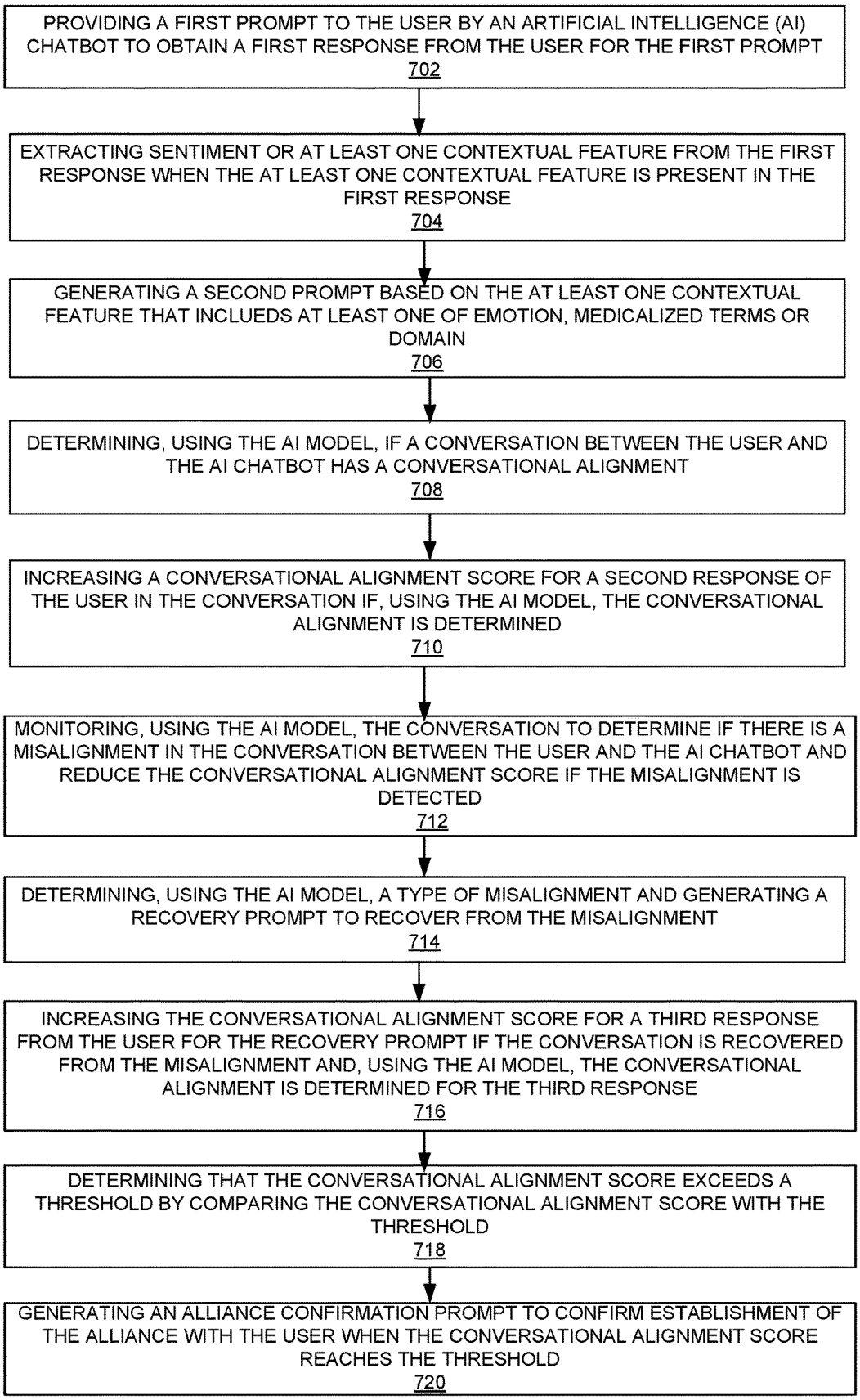

PROVIDING A FIRST PROMPT TO THE USER BY AN ARTIFICIAL INTELLIGENCE (AI) CHATBOT TO OBTAIN A FIRST RESPONSE FROM THE USER FOR THE FIRST PROMPT
702
EXTRACTING SENTIMENT OR AT LEAST ONE CONTEXTUAL FEATURE FROM THE FIRST RESPONSE WHEN THE AT LEAST ONE CONTEXTUAL FEATURE IS PRESENT IN THE FIRST RESPONSE
704
GENERATING A SECOND PROMPT BASED ON THE AT LEAST ONE CONTEXTUAL FEATURE THAT INCLUEDS AT LEAST ONE OF EMOTION, MEDICALIZED TERMS OR DOMAIN
706
DETERMINING, USING THE AI MODEL, IF A CONVERSATION BETWEEN THE USER AND THE AI CHATBOT HAS A CONVERSATIONAL ALIGNMENT
708
INCREASING A CONVERSATIONAL ALIGNMENT SCORE FOR A SECOND RESPONSE OF THE USER IN THE CONVERSATION IF, USING THE AI MODEL, THE CONVERSATIONAL ALIGNMENT IS DETERMINED
710
MONITORING, USING THE AI MODEL, THE CONVERSATION TO DETERMINE IF THERE IS A MISALIGNMENT IN THE CONVERSATION BETWEEN THE USER AND THE AI CHATBOT AND REDUCE THE CONVERSATIONAL ALIGNMENT SCORE IF THE MISALIGNMENT IS DETECTED
712
DETERMINING, USING THE AI MODEL, A TYPE OF MISALIGNMENT AND GENERATING A RECOVERY PROMPT TO RECOVER FROM THE MISALIGNMENT
714
INCREASING THE CONVERSATIONAL ALIGNMENT SCORE FOR A THIRD RESPONSE FROM THE USER FOR THE RECOVERY PROMPT IF THE CONVERSATION IS RECOVERED FROM THE MISALIGNMENT AND, USING THE AI MODEL, THE CONVERSATIONAL ALIGNMENT IS DETERMINED FOR THE THIRD RESPONSE
716
DETERMINING THAT THE CONVERSATIONAL ALIGNMENT SCORE EXCEEDS A THRESHOLD BY COMPARING THE CONVERSATIONAL ALIGNMENT SCORE WITH THE THRESHOLD
718
GENERATING AN ALLIANCE CONFIRMATION PROMPT TO CONFIRM ESTABLISHMENT OF THE ALLIANCE WITH THE USER WHEN THE CONVERSATIONAL ALIGNMENT SCORE REACHES THE THRESHOLD
720
700

FIG. 7

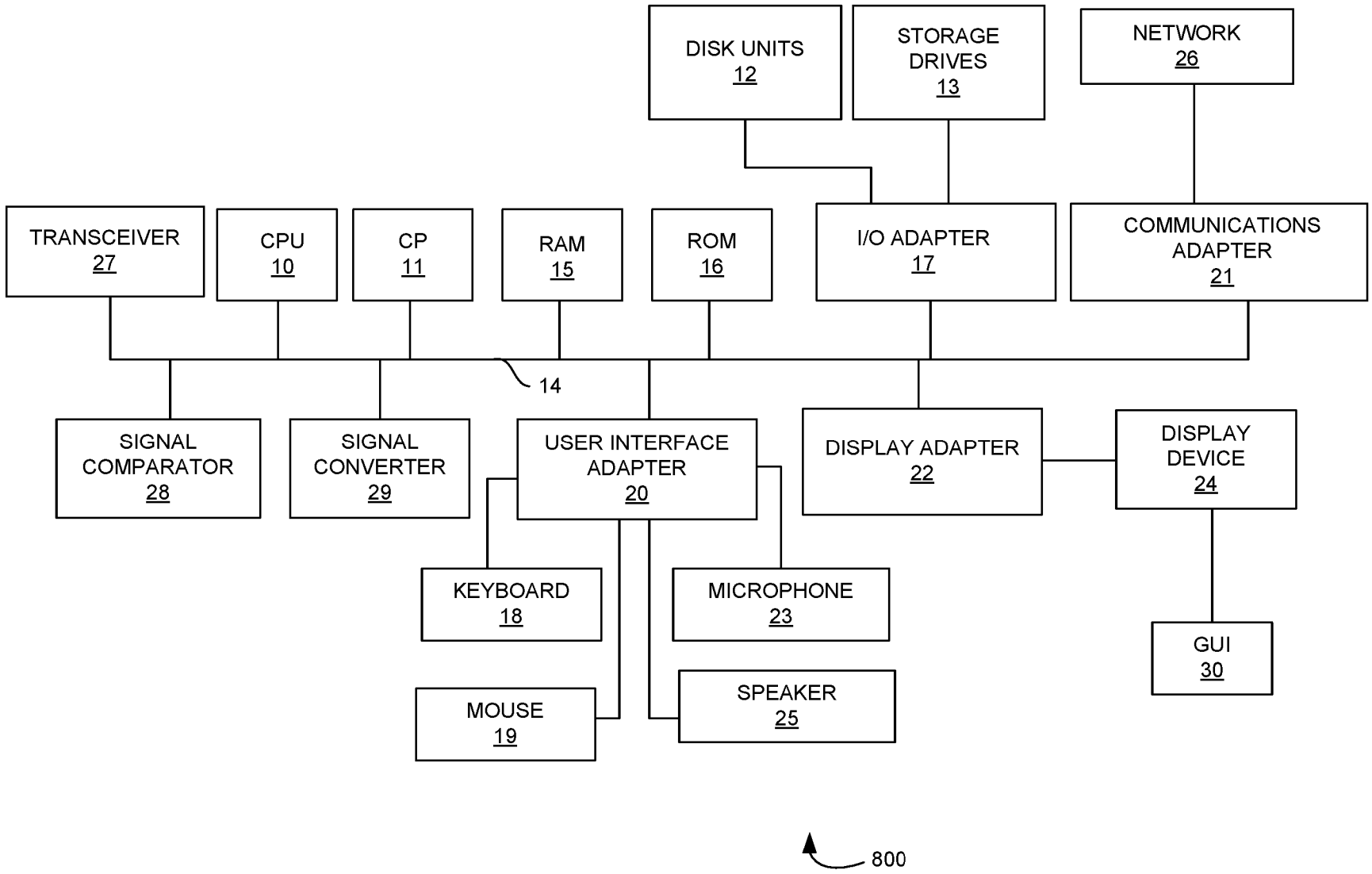

DISK UNITS
12
STORAGE
DRIVES
13
NETWORK
26
TRANSCEIVER
27
CPU
10
CP
11
RAM
15
ROM
16
I/O ADAPTER
17
COMMUNICATIONS
ADAPTER
21
14
SIGNAL
COMPARATOR
28
SIGNAL
CONVERTER
29
USER INTERFACE
ADAPTER
20
DISPLAY ADAPTER
22
DISPLAY
DEVICE
24
KEYBOARD
18
MICROPHONE
23
GUI
30
MOUSE
19
SPEAKER
25
800

FIG. 8

SYSTEM AND METHOD FOR MONITORING AND IMPROVING CONVERSATIONAL ALIGNMENT TO DEVELOP AN ALLIANCE BETWEEN AN ARTIFICIAL INTELLIGENCE (AI) CHATBOT AND A USER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application claims priority to pending U.S. provisional patent application No. 63/290,045 filed on Dec. 15, 2021, the complete disclosures of which, in their entirety, are hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] Embodiments of this disclosure generally relate to an artificial intelligence (AI) chatbot, and more particularly, to a system and method for monitoring and improving conversational alignment to develop an alliance between the artificial intelligence (AI) chatbot and a user.

BACKGROUND

[0003] A chatbot is a computer program that simulates and processes human conversation to allow a human to interact with a digital device as if it were a real person. Conversational chatbots or digital assistants leverage natural-language understanding (NLU), natural language processing (NLP), and machine learning (ML) to understand and respond to a user's requests or queries, learn a user's preferences over time, provide recommendations, and even anticipate needs. Artificial intelligence chatbots are chatbots trained to have human-like conversations using a process known as natural language processing (NLP). With NLP, the AI chatbot can interpret human language as it is written, which enables it to carry on a conversation with the user without the need of any human intervention in normal course. Further, AI chatbot software can understand language outside of pre-programmed commands and provide a response based on existing data. This allows the users to lead the conversation, voicing their intent in their own words. However, if a demand is made that extends beyond the chatbot's capabilities, it might struggle.

[0004] Further, chatbots are typically not capable of empathizing with or emotionally relating to the user. Hence, it is difficult for a chatbot to build sufficient trust with a user for the user to open up and share their thoughts or feelings with the chatbot. For example, if the user feels low, upset, or frustrated, and expresses his/her feelings to the chatbot either directly or indirectly during the conversation, the chatbot may seem to lack empathy even if it correctly understands what the user is saying, because the chatbot may not be explicitly designed to give empathetic responses and a 'lending ear' to the user. The chatbot will move on with the conversation without showing any empathy, and even if its response is the 'transactionally correct' response, the lack of empathy could substantially reduce trust in the chatbot and its ability to help the user.

[0005] This can lead to the user feeling more frustrated, exiting the conversation, and/or being dissuaded from engaging with the chatbot in the future. On the other hand, if the chatbot is able to respond in a manner that indicates understanding and empathy, it will enhance trust. When there is trust, a user is more likely to engage with the chatbot, and even respond positively to suggestions or recommendations made by the chatbot. Hence, there remains a challenge for the chatbot to make the user feel heard and understood, and win the user's trust, which would encourage the user to respond positively to suggestions made by the chatbot. This also suggests the need for an ability of the chatbot to recover from any situations when the user's trust in it seems to have reduced.

SUMMARY

[0006] In view of the foregoing, embodiments herein provide a processor-implemented method for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user. The method includes providing a first prompt to the user by the AI chatbot to obtain a first response from a user device associated with the user for the first prompt. The first response is at least one of a text input or a voice input. The method includes extracting sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response. Extracting the at least one contextual feature comprises detecting that the first response of the user includes at least one of emotion, medicalized terms or domain, wherein (i) the emotion is detected using an emotion detecting artificial intelligence (AI) model, (ii) the medicalized terms are detected using a medicalized term detecting AI model, and (iii) the domain is detected using a domain detecting AI model. The method includes generating a second prompt based on the at least one contextual feature includes at least one of the emotion, the medicalized terms or the domain. The method includes determining, using the AI model, if a conversation between the user and the AI chatbot has a conversational alignment. The conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot and agreeing to suggestions or interpretations made by the AI chatbot. The method includes increasing a conversational alignment score for a second response of the user in the conversation if, using the AI model, the conversational alignment is determined. The method includes monitoring, using the AI model, the conversation to determine if there is a misalignment in the conversation between the user and the AI chatbot and reduce the conversational alignment score if the misalignment is detected. The method includes determining, using the AI model, a type of misalignment and generating a recovery prompt to recover from the misalignment. The method includes increasing the conversational alignment score for a third response from the user for the recovery prompt if the conversation is recovered from the misalignment and, using the AI model, the conversational alignment is determined for the third response. The method includes determining that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold. The method includes generating an alliance confirmation prompt to confirm establishment of the alliance with the user when the conversational alignment score reaches the threshold.

[0007] In some embodiments, the method includes if a plurality of contextual features are detected in each response of the user, prioritizing the plurality of contextual features to decide which direction to take the conversation. The plurality of contextual features may be prioritized as (i) the at least one medicalized term, (ii) domain, and (iii) emotion

[0008] In some embodiments, prompts are generated based on a contextual feature that has highest priority among

the plurality of contextual features, wherein the contextual feature that has the highest priority is identified by prioritizing the plurality of contextual features in a decreasing order.

[0009] In some embodiments, the method includes creating an intent for each type of misalignment for intent recognition by providing representative patterns for each type of misalignment. In some embodiments, the type of misalignment is determined using an intent recognition AI model. The type of misalignment may be selected from at least one of affirmation, confusion, disagreement, dissatisfaction, lack of trust, refusal, or uncertainty expressed by the user to the AI chatbot.

[0010] In some embodiments, at least one of the affirmation, the confusion, the disagreement, the dissatisfaction, the lack of trust, the refusal or uncertainty expressed by the user to the AI chatbot is detected using a classifier.

[0011] In some embodiments, the type of misalignment is classified using a plurality of machine learning models. In some embodiments, the plurality of machine learning models are trained based on training data that comprises examples of user text labeled as true, and the user text labeled as false.

[0012] In some embodiments, the method includes determining closest semantic and syntactic match for each response received from the user among all the representative patterns provided.

[0013] In some embodiments, if a confidence score for the matching pattern with a highest confidence score is above an intent matching threshold, a response received from the user is determined to correspond to the intent that the matching pattern represents.

[0014] In some embodiments, the recovery prompt is generated based on the type of misalignment determined.

[0015] In some embodiments, the conversational alignment score is updated after each response received from the user at the AI chatbot during the conversation and indicates strength of the conversational alignment formed between the AI chatbot and the user.

[0016] In some embodiments, the method includes vectorizing the text input received from the user to convert the user text into a numerical representation to use the numerical representation corresponding to the user text for training an AI model and for inferencing. In some embodiments, the text input is vectorized using frequency-based techniques or semantics-based techniques.

[0017] In some embodiments, the prompts are generated based on predefined base prompts that are written by conversation designers and parameterized with the user's context to personalize them. In some embodiments, the predefined base prompts are stored in a database of a server.

[0018] In some embodiments, the prompts are provided to the user through the AI chatbot until the conversational alignment score exceeds the threshold.

[0019] In some embodiments, at least one of the open-ended prompts or the closed-ended prompts are alternately provided to the user based on the conversational alignment score.

[0020] In some embodiments, the method includes recommending at least one intervention to the user when the alliance has been confirmed using the alliance confirmation prompt in which the user agrees to try out an intervention.

[0021] In one aspect, one or more non-transitory computer readable storage mediums storing one or more sequences of instructions, which when executed by one or more processors, causes a method for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user by performing the steps. The method includes providing a first prompt to the user by the AI chatbot to obtain a first response from a user device associated with the user for the first prompt. The first response is at least one of a text input or a voice input. The method includes extracting sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response. Extracting the at least one contextual feature comprises detecting that the first response of the user includes at least one of emotion, medicalized terms or domain, wherein (i) the emotion is detected using an emotion detecting artificial intelligence (AI) model, (ii) the medicalized terms are detected using a medicalized term detecting AI model, and (iii) the domain is detected using a domain detecting AI model. The method includes generating, using an AI model, a second prompt based on the at least one contextual feature includes at least one of the emotion, the medicalized terms or the domain. The method includes determining, using the AI model, if a conversation between the user and the AI chatbot has a conversational alignment. The conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot and agreeing to suggestions or interpretations made by the AI chatbot. The method includes increasing, using the AI model, a conversational alignment score for a second response of the user in the conversation if the conversational alignment is determined. The method includes monitoring, using the AI model, the conversation to determine if there is a misalignment in the conversation between the user and the AI chatbot and reduce the conversational alignment score if the misalignment is detected. The method includes determining, using the AI model, a type of misalignment and generating a recovery prompt to recover from the misalignment. The method includes increasing, using the AI model, the conversational alignment score for a third response from the user for the recovery prompt if the conversation is recovered from the misalignment and the conversational alignment is determined for the third response. The method includes determining that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold. The method includes generating an alliance confirmation prompt to confirm establishment of the alliance with the user when the conversational alignment score reaches the threshold.

[0022] In some embodiments, the method includes if a plurality of contextual features are detected in each response of the user, prioritizing the plurality of contextual features to decide which direction to take the conversation. The plurality of contextual features may be prioritized as (i) the at least one medicalized term, (ii) domain, and (iii) emotion

[0023] In some embodiments, prompts are generated based on a contextual feature that has highest priority among the plurality of contextual features, wherein the contextual feature that has the highest priority is identified by prioritizing the plurality of contextual features in a decreasing order.

[0024] In some embodiments, the method includes creating an intent for each type of misalignment for intent recognition by providing representative patterns for each type of misalignment. In some embodiments, the type of

misalignment is determined using an intent recognition AI model. The type of misalignment may be selected from at least one of affirmation, confusion, disagreement, dissatisfaction, lack of trust, refusal or uncertainty expressed by the user to the AI chatbot.

[0025] In the second aspect, a system for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user is provided. The system includes a device processor and a non-transitory computer readable storage medium storing one or more sequences of instructions, which when executed by the device processor, causes a method by performing the steps. The method includes providing a first prompt to the user by the AI chatbot to obtain a first response from the user for the first prompt. The first response is at least one of a text input or a voice input. The method includes extracting sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response. Extracting the at least one contextual feature comprises detecting that the first response of the user includes at least one of emotion, medicalized terms or domain, wherein (i) the emotion is detected using an emotion detecting artificial intelligence (AI) model, (ii) the medicalized terms are detected using a medicalized term detecting AI model, and (iii) the domain is detected using a domain detecting AI model. The method includes generating, using an AI model, a second prompt based on the at least one contextual feature includes at least one of the emotion, the medicalized terms or the domain. The method includes determining, using the AI model, if a conversation between the user and the AI chatbot has a conversational alignment. The conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot and agreeing to suggestions or interpretations made by the AI chatbot. The method includes increasing, using the AI model, a conversational alignment score for a second response of the user in the conversation if the conversational alignment is determined. The method includes monitoring, using the AI model, the conversation to determine if there is a misalignment in the conversation between the user and the AI chatbot and reduce the conversational alignment score if the misalignment is detected. The method includes determining, using the AI model, a type of misalignment and generating a recovery prompt to recover from the misalignment. The method includes increasing, using the AI model, the conversational alignment score for a third response from the user for the recovery prompt if the conversation is recovered from the misalignment and the conversational alignment is determined for the third response. The method includes determining that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold. The method includes generating an alliance confirmation prompt to confirm establishment of the alliance with the user when the conversational alignment score reaches the threshold.

[0026] These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The embodiments herein will be better understood from the following detailed description with reference to the drawings, in which:

[0028] FIG. **1** is a block diagram that illustrates a system that monitors and improves conversational alignment between an artificial intelligence (AI) chatbot and a user according to some embodiments herein;

[0029] FIG. **2** is a block diagram of a server of FIG. **1** according to some embodiments herein;

[0030] FIGS. **3**A-**3**B are mock-up screenshots of user interfaces that illustrate a conversation between the AI chatbot and the user without misalignments according to some embodiments herein;

[0031] FIGS. **4**A-**4**C are mock-up screenshots of user interfaces that illustrate a conversation between the AI chatbot and the user with one or more misalignments according to some embodiments herein;

[0032] FIG. **4**D is an exemplary graphical representation of the conversational alignment score with the one or more misalignments according to some embodiments herein;

[0033] FIGS. **5**A-**5**C are mock-up screenshots of user interfaces that illustrate a conversation between the user and the AI chatbot according to some embodiments herein;

[0034] FIG. **6** is a flowchart that illustrates a method for monitoring a conversational alignment and establishing an alliance between the artificial intelligence (AI) chatbot and the user by the server through the AI chatbot according to some embodiments herein;

[0035] FIG. **7** is a flow diagram that illustrates a method for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user according to some embodiments herein; and

[0036] FIG. **8** is a block diagram of a schematic diagram of a device used in accordance with embodiments herein.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0037] The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments.

[0038] As used herein, the following terms and phrases shall have the meanings set forth below. Unless defined otherwise, all technical terms used herein have the same meaning as commonly understood to one of ordinary skill in the art. The singular forms “a”, “an”, and “the” include plural reference unless the context clearly dictates otherwise.

Definitions

[0039] The term "open-ended prompt" is a question that cannot be answered with a "yes" or "no" response, or with one of a finite set of responses. Open-ended prompts are phrased as a statement which requires a longer response.

[0040] The term "closed-ended prompt" is a question that could potentially be answered with a "yes" or "no" response, or with one of a finite set of responses. However, they are phrased in such a way that the response could be a longer free text response as well.

[0041] Referring now to the drawings, and more particularly to FIGS. **1** through **8**, where similar reference characters denote corresponding features consistently throughout the figures, there are shown preferred embodiments.

[0042] FIG. **1** is a block diagram **100** that illustrates a system that monitors and improves a conversational alignment between an artificial intelligence (AI) chatbot **110** and a user **102** according to some embodiments herein. The block diagram **100** includes a user device **104** associated with the user **102**, a network **106**, and a server **108** that includes an artificial intelligence (AI) model **112**. In some embodiments, the user device **104** includes the Artificial Intelligence (AI) chatbot **110**. In some embodiments, the user device **104**, without limitation, may be selected from a mobile phone, a Personal Digital Assistant (PDA), a tablet, a desktop computer, or a laptop.

[0043] The user device **104** may communicate with the server **108** through the network **106**. In some embodiments, the network **106** is a wired network, a wireless network, or a combination of a wired network and a wireless network. In some embodiments, the network **106** is the Internet. The AI model **112** of the server **108** detects the sentiment (examples of sentiment types include positive, negative, and neutral) of a first response of the user **102**. The first response is at least one of a text input or a voice input. In some embodiments, before training the AI model **112**, the text input received from the user **102** is preprocessed that include steps such as spell correction, expansion of contractions, removal of stop words, stemming and lemmatization. The user text then needs to be vectorized so that the text is converted into a numerical representation which the AI model **112** can work with. In some embodiments, the text input is vectorized using frequency-based techniques such as Bag of Words (BoW), CountVectorizer, or TF-IDF, or semantics-based techniques such as word embeddings (eg., Word2Vec, GloVe, etc.). In some embodiments, pre-trained models from SpaCy, fastText, etc. can also be used for word embeddings. In some embodiments, the AI model **112** converts the first response or a second response received from the user **102** into the text input if the first response or the second response is the voice input using Natural Language Processing (NLP).

[0044] The AI chatbot **110** starts a conversation with the user **102** with an empathetic statement based on the sentiment of the user **102**. As the user **102** responds to the empathetic statement and the conversation begins, the server **108** starts gathering context from the user **102**. The AI model **112** dynamically generates prompts such as a first prompt, a second prompt, an alliance confirmation prompt, and a recovery prompt based on context gathered from the user **102**. In some embodiments, the prompts are at least one of open-ended prompts or closed-ended prompts. In some embodiments, the context gathered from the user **102** includes extraction of one or more contextual features.

[0045] In some embodiments, the one or more contextual features are used to assist the chatbot **110** to understand and respond to the user **102**. In some embodiments, the one or more contextual features that are extracted from a user response include suicidal thoughts, mentions of self-harm or abuse, emotions, domains (i.e., a topic of conversation), medicalized terms, entities, agreement with the chatbot **110**, dissatisfaction or disagreement with the chatbot **110**, uncertainty, sentiment, confusion, and lack of trust in the AI chatbot **110**. In some embodiments, if a contextual feature is not extracted from the user message, the AI model **112** automatically provides at least one of an open-ended prompt or a closed-ended prompt to obtain more context from the user **102**. In some embodiments, if only one of the contextual features is identified in the user message, the AI model **112** may generate at least one of the open-ended or the closed-ended prompt based on that one contextual feature. In some embodiments, if the one or more contextual features are identified in the gathered context, the AI model **112** prioritizes the one or more contextual features based on an importance of the one or more features. The AI model **112** may generate at least one of the open-ended or the closed-ended prompt based on the most important contextual feature identified in the user text. In some embodiments, the contextual features are prioritized in decreasing order of importance as suicidal thoughts, mentions of self-harm or abuse, confusion, dissatisfaction, lack of trust or disagreement with the AI chatbot **110**, uncertainty, medicalized terms, domains, emotions, agreement with the AI chatbot **110**, and sentiment.

[0046] In some embodiments, the AI model **112** provides at least one of the open-ended or closed-ended prompts to the user **102** through the AI chatbot **110** to gather more context from the user **102**. In some embodiments, the AI model **112** provides at least one of the open-ended or closed-ended prompts to the user **102** through the AI chatbot **110** until an alliance is confirmed to be established between the user **102** and the AI chatbot **110**. In some embodiments, the AI chatbot **110** may not wait for the alliance to be established before recommending a solution to the user **102** if the user message indicates an urgent need that must be immediately addressed.

[0047] In some embodiments, the server **108** includes one or more AI models to detect and extract one or more relevant conversational features of the user text. The one or more relevant features correspond to any aspect of the user message that are used to assist the AI chatbot **110** to understand and respond to the user **102** better so that the user **102** feels heard and understood. In some embodiments, detection of the one or more relevant conversational features include detection of whether the user **102** is agreeing (e.g., "Yes, that's right.") or disagreeing with the chatbot **110** (e.g., "I don't think so"), whether the user **102** is in distress and might need immediate help (e.g., "I'm having a panic attack"), confused (e.g., "What do you mean?"), unhappy with the chatbot **110** (e.g., "You're not even helping!"), uncertain about how to respond (e.g., "Umm . . . I don't really know"), lacking trust in the AI chatbot **110** (e.g., "I don't think you can help me"), etc. In addition, the AI model **112** extracts one or more contextual features that are used to assist the chatbot **110** to respond to the user **102** better. The contextual features may include a domain, e.g., what the user **102** is talking about (relationship, education, health, money, politics, sports, etc.); entity, e.g., who the user **102**

is talking about (self, family member, coworker, friend, etc.); sentiment, i.e., whether the tone of the user **102**'s message is positive, negative or neutral; emotion, e.g., how the user **102** is feeling (sad, angry, happy, frustrated, scared, etc.); among other things the user **102** might mention (activities, medicalized terms, events, etc.).

[0048] The AI models may be trained on training data obtained from multiple sources, such as user messages from conversations in the past with any Personal Identifiable Information (PII) replaced by synthetic data, manually created examples (i.e., synthetic statements manually created by data scientists based on the expected response of the user **102**), third-party public datasets, e.g., Kaggle, and Web scraping, e.g., from Reddit, Twitter by Twitter, Inc, news websites, etc.

[0049] In some embodiments, the open-ended prompts or the closed-ended prompts are generated based on predefined base prompts written by conversation designers. These base prompts are parameterized with the user's context to personalize them. For example, an open-ended prompt could comprise a context-based empathetic statement followed by a predefined base prompt like. "How did (context) make you feel?". Thus, if the user **102** talks about a conflict with a colleague, the open-ended prompt would comprise an empathetic statement based on this context like "I understand things are not going well with your colleague at the moment. ", followed by the contextualized base prompt like "How did this conflict make you feel?". In some embodiments, the predefined base prompts are stored in the server **108**. In some embodiments, the open-ended prompts or the closed-ended prompts are provided alternately to the user **102** based on a conversational alignment score.

[0050] The AI model **112** continuously monitors the conversation between the user **102** and the AI chatbot **110** to determine if the conversation between the user **102** and the AI chatbot **110** has conversational alignment. The conversational alignment is an alignment with respect to the user **102** sharing more context or agreeing with the AI chatbot **110** when prompted by the AI chatbot **110**. For example, if the AI chatbot **110** provides the open-ended prompt, (e.g., "What bothered you the most about this situation?") or the closed-ended prompt (e.g., "Do you sometimes wish that you had more control over how others act?"), the user **102** may share more (e.g., "That I am not considered important by my family") or agree with the chatbot **110** (e.g., "Yes, I think so."), thus indicating conversational alignment. In some embodiments, the AI model **112** determines a misalignment by the user's confusion, (e.g., "What do you mean by that?"), dissatisfaction (e.g., "You don't understand me!"), disagreement (e.g., "No, that's not what I meant"), lack of trust (e.g., "A bot can't help me"), or uncertainty (e.g., "I'm not sure").

[0051] The AI model **112** maintains the conversational alignment score, which indicates the strength of the alliance formed between the AI chatbot **110** and the user **102**. The conversational alignment score is updated after each message exchange based on the conversation of the user **102** with the AI chatbot **110**. For example, the conversational alignment score is increased in case of the conversational alignment, i.e., if the user **102** goes with the flow, responds to the AI chatbot **110** promptly by sharing more, or agrees with the AI chatbot **110**. The conversational alignment score is decreased when the AI model **112** detects a lack of conversational alignment, i.e., when the user **102** doesn't agree with the AI chatbot **110**, is confused, dissatisfied, or unsure about how to respond to the AI chatbot **110**.

[0052] If the conversation between the user **102** and the AI chatbot **110** is not aligned, the server **108** reduces the conversational alignment score and attempts to recover from this setback by addressing the user's concerns empathetically and bringing the user **102** back to the conversation. For example, if the AI chatbot **110** provides the open-ended prompt, e.g., "How did that make you feel?" and the user **102** expresses uncertainty, e.g., "I don't really know . . . can't say . . . ", the AI chatbot **110** will respond empathetically, e.g., "That's also okay. Sometimes our feelings go into hiding for a bit." and bring the user **102** back to the conversation, e.g., "Do you think there is anything in this situation that you can change?". Another example could be when the user **102** expresses dissatisfaction, e. g., "I don't think you can help me", the AI chatbot **110** may respond to the user **102**, e.g., "I'm trying my best to understand you better. Try venting to me, it will help.".

[0053] The AI model **112** continuously monitors the conversational alignment score and compares the conversational alignment score to a threshold after every message exchange and score update. In some embodiments, the threshold for the conversational alignment score is determined based on data analysis to determine which value results in best performance over a validation dataset. When the conversational alignment score exceeds the threshold, the server **108** dynamically provides an alliance confirmation prompt to the user **102** to confirm the establishment of the alliance and checks if the user **102** is now ready to accept an intervention, e.g., "Thanks for sharing that with me. I have something that could help you manage this better. Would you like to try it out?". In some embodiments, when the alliance is established, the alliance may be similar to a therapeutic alliance between a human therapist and the user **102**. In some embodiments, when the alliance is established and confirmed by the user **102**, the AI chatbot **110** may recommend some interventions or solutions to the user **102**. In some embodiments, a recommended activity is a therapeutic intervention, e.g., a breathing exercise, a physical activity, writing down and reframing thoughts, etc.

[0054] FIG. **2** is a block diagram **200** of the server **108** of FIG. **1** according to some embodiments herein. The server **108** includes a database **200**, a sentiment detecting module **202**, a context gathering module **204**, a critical situation detecting module **205**, a prompt generating module **211**, a conversational alignment monitoring module **212**, a conversational alignment score monitoring module **218**, an alliance confirmation module **220**, and an intervention suggesting module **222**. The context gathering module **204** includes an emotion detecting AI model **206**, a medicalized term detecting AI model **208**, and a domain detecting AI model **210**. The conversational alignment monitoring module **212** includes an alignment detecting module **214** and a misalignment detecting module **216**. The sentiment detecting module **202** sentiment detects sentiment (positive, negative, or neutral) of the user's opening text and starts a conversation with the user **102** with an empathetic statement based on the sentiment of the user **102**.

[0055] The context gathering module **204** extracts one or more contextual features from the user text. In some embodiments, contextual features are extracted using the emotion detecting AI model **206**, the medicalized term detecting AI model **208**, and the domain detecting AI model

210. In some embodiments, the emotion detecting AI model **206** detects emotion words such as "anxious", "scared", and "restless" expressed by the user **102** to direct the conversation accordingly. In some embodiments, the medicalized term detecting AI model **208** detects medicalized terms such as panic attack, depression, and diabetes in the user text.

[0056] In some embodiments, if the medicalized term detecting AI model **208** detects one or more medicalized terms, the medicalized term detecting AI model **208** then checks if the user **102** is mentioning those terms with respect to themselves, and if the user **102** is in distress and needs immediate help. The domain detecting AI model **210** extracts a topic of the conversation such as relationships, work, education, and abuse from the user text. In some embodiments, if multiple contextual features (e.g., emotions, domains, medicalized terms) are detected in the context gathered from the user **102**, they are prioritized in the following order to decide which direction to take the conversation in: (i) medicalized terms, (ii) domain, and (iii) emotion. In some embodiments, the relevant pieces of contextual information extracted from the user's texts are stored in the database **200** of the server **108**. For example, if the medicalized term is detected, the open-ended prompt, e.g., "What does a professional have to say about this?" or "What's the hardest part about {detected_medicalized_term}?" is provided to the user **102**. Similarly, if the domain is detected, the open-ended prompt, e.g., "What's the hardest part about that?" is given, and in case of the detected emotion, a prompt, e.g., "Tell me more about this feeling." is given to the user **102**.

[0057] The prompt generating module **211** dynamically generates at least one of open-ended or closed-ended prompts based on relevant pieces of the one or more contextual features extracted from the user's texts to make the user **102** feel heard and understood, and to obtain further context from the user **102**.

[0058] In some embodiments, the open-ended prompts or the closed-ended prompts are customized in such a way that the open-ended prompts or the closed-ended prompts are specific to a type of context detected to fit a variety of situations. In some embodiments, the open-ended prompts or the closed-ended prompts are generated by the prompt generating module **211** based on predefined base prompts that are written by conversation designers. These base prompts are parameterized with the user's context to personalize them. For example, a predefined base prompt may be "(empathetic statement based on context). How did (context) make you feel?". If the user **102** talks about conflict with a colleague, the predefined base prompt will be personalized like "I understand things are not going well with your colleague at the moment. How did this conflict make you feel?" In some embodiments, the predefined base prompts are enabled to ensure that generated prompts are safe and do not trigger the user **102** by inadvertently saying something inappropriate.

[0059] In some embodiments, the prompt generating module **211** offers an empathetic statement followed by at least one of open-ended or closed-ended prompts to the user **102** based on the extracted context. For example, if a medicalized term such as 'depression' is detected, an empathetic open-ended prompt such as "That can feel like an exhausting battle within. What's the most difficult part about depression?" is given to the user **102**. Similarly, if a domain like 'education failure' is detected, an empathetic open-ended prompt such as "That sounds hard. While success gives us energy, failure helps us grow. Both are a part of making you the awesome person you are. I'm certain you will bounce back no matter what happens. As challenging as this is, what would you say is the most difficult part of it for you right now?" is given, and in case of a detected emotion like 'sadness', an empathetic open-ended prompt like "I understand you're feeling sad. It's only natural to feel this way. Tell me more about this feeling." is shown.

[0060] In some embodiments, the prompt generating module **211** offers prompts in such a way that while they are specific to a type of context detected, they are still generic enough to fit a variety of situations. For example, a generic open-ended prompt designed for the "fear" emotion could be "Breathe deeply. Everything that you need to cope with your fear is within you. Tell me more about this feeling." While this prompt is tailored to a specific emotion, it is generic enough to fit any situation where the user might be feeling scared. In some embodiments, if no contextual feature is extracted from the user text, a generic open-ended or closed-ended prompt like "I hear you. What's the hardest part about that?" is offered which fits a variety of situations even though the exact context is unknown.

[0061] In some embodiments, based on the user's response to the open-ended prompt and the further context derived from it, if any, the prompt generating module **211** offers a follow-up closed-ended question like "Is there something that has been already helping you cope?", "Are there some things in your control that you can change?", or "Do you often struggle with this feeling?" through the AI chatbot **110**.

[0062] The conversational alignment monitoring module **212** continuously monitors the conversation between the user **102** and the AI chatbot **110** to determine if the user's response is aligned with the AI chatbot's **110** message or not. In some embodiments, this monitoring includes checking for alignment, i.e., whether the user **102** shared any more context or agreed with the AI chatbot **110** and misalignment, i.e., whether the user **102** expressed confusion, disagreement, dissatisfaction or uncertainty. The conversational alignment monitoring module **212** maintains a conversational alignment score for the conversation and updates the conversational alignment score based on whether the conversational alignment between the user **102** and the AI chatbot **110** is identified or not after every message exchange.

[0063] If the alignment detecting module **214** detects that the conversation between the user **102** and the AI chatbot **110** is aligned (e.g., the user **102** shared more context when prompted by the AI chatbot **110**, or agreed with the AI chatbot **110**), the conversational alignment score is increased. For example, the alignment detecting module **214** may increase the conversational alignment score if the AI chatbot **110** asks "What's bothering you?" and the user **102** responds with "I had a fight with my mom." since the user **102** shared more when prompted by the AI chatbot **110**.

[0064] If the misalignment detecting module **216** detects that there is misalignment in the conversation between the user **102** and the AI chatbot **110** (e.g., the user **102** expressed confusion, disagreement, dissatisfaction or uncertainty when prompted by the AI chatbot **110**), the conversational alignment score is decreased. The misalignment detecting module **216** determines a type of misalignment and generates a recovery prompt to recover from the misalignment. In some

embodiments, the recovery prompt is generated based on the type of misalignment determined.

[0065] In some embodiments, the misalignment detecting module **216** creates an intent for each type of misalignment for intent recognition by providing representative patterns for each type of misalignment. The type of misalignment may be determined using an intent recognition AI model. The type of misalignment may be selected from at least one of confusion, disagreement, dissatisfaction, lack of trust, refusal or uncertainty expressed by the user **102** to the AI chatbot **110**.

[0066] For example, the misalignment is detected if the user **102** may be confused as to what the AI chatbot **110** is saying or asking them, e.g., "What do you mean by that?", "Can you explain?", "I didn't get you", "Come again?" and "I don't understand". The misalignment is detected if the user **102** may disagree with what the AI chatbot **110** said, e.g., "No, that's not what I meant", "Not really", "That's not true", "I don't think so", etc. The misalignment is detected if the user **102** may be unsure about how to respond to the AI chatbot **110**, e.g., "I'm not sure", "I don't really know", "Umm . . . can't say", "Dunno", "Idk", etc. The misalignment is detected if the user **102** may be dissatisfied or unhappy with what the AI chatbot **110** is asking them or how the AI chatbot **110** is responding to them, e.g., "You don't understand me!", "Are you even listening?", "You're just a stupid ai", "I don't want to do this", "I already told you this", "Stop repeating yourself", "You don't really understand what I'm saying, do you?", etc.

[0067] The misalignment is detected if the user **102** may not trust the AI chatbot **110** enough, e.g., "You can't help me", "You wouldn't understand", "You are just a bot. How can you help me?", "I don't think you can make me feel any better", etc. The misalignment is detected if the user **102** may refuse to do what the AI chatbot **110** suggests them to do, e.g., "I don't want to", "I can't", "Not really", "No, can we do something else?", etc.

[0068] In some embodiments, the closest semantic and syntactic match among all the representative patterns provided for each intent is determined for each response received from the user **102**. In some embodiments, if the confidence score for the matching pattern with a highest confidence score is above an intent matching threshold, a response received from the user **102** is determined to correspond to the intent that the matching pattern represents. For example, if the user text "Can you even understand me?" finds the best match in the pattern "You don't understand me!" with a confidence score higher than the intent matching threshold, then the user text can be detected as a misalignment of the dissatisfaction type because the matching pattern belongs to that intent.

[0069] In some embodiments, if the misalignment detecting module **216** detects a misalignment, the conversational alignment score is reduced. For example, the misalignment detecting module **216** may reduce the conversational alignment score if the AI chatbot **110** asks "What's bothering you?" and the user **102** expresses uncertainty by responding with "I don't really know".

[0070] When the misalignment detecting module **216** detects a decrease in alliance (the user **102** expresses confusion, disagreement, dissatisfaction or uncertainty when prompted by the AI chatbot **110**), the prompt generating module **211** attempts to recover from this setback by addressing the user's concerns empathetically and bringing the user **102** back to the conversation. For example, if the AI chatbot **110** offered an open-ended prompt like "How did that make you feel?" and the user **102** expresses lack of trust or dissatisfaction by saying "You're just a bot. You can't help me.", the AI chatbot **110** may try to recover from this decrease in alliance by responding with "I'm trying my best to understand you better. Try venting to me, it will help.".

[0071] The conversational alignment score monitoring module **218** continuously monitors the conversational alignment score to determine if the conversational alignment score exceeds a threshold and determines if the user **102** feels heard and understood by the AI chatbot **110** and is ready to accept an intervention or a recommendation. The intervention suggesting module **222** checks the user's readiness to use an intervention. At this point, the alliance confirmation module **220** dynamically provides an alliance confirmation prompt that is generated by the prompt generating module **211** to confirm the establishment of the alliance with the user **102** when the conversational alignment score exceeds the threshold. This closed-ended prompt could look like "I may have something to help you manage your feelings around this. Would you like to give it a try?" If the user **102** accepts this "handshake", the alliance is confirmed to have been formed.

[0072] In some embodiments, if the context gathering module **204** detects that the user **102** is in distress and needs immediate help, the critical situation detection module **205** takes over and immediately offers a suitable intervention to the user **102** based on the context gathered without waiting for an alliance to be formed through a conversational alliance scoring mechanism.

[0073] In some embodiments, the AI chatbot **110** is a digital assistant for mental health. FIGS. **3A-3B** are mock-up screenshots of user interfaces that illustrate a conversation between the AI chatbot **110** and the user **102** without misalignments according to some embodiments herein. In FIG. **3A** of a user interface **300**, the AI chatbot **110** starts a conversation with the user **102** with an empathetic statement **302**, e.g., "Alex, it's nice to see you! How's your day going so far?". The user **102** may respond with **304** to the empathetic statement **302** provided by the AI chatbot **110**, e.g., "Don't even ask." The sentiment detecting module **202** detects sentiment of the user's response **304**. Based on the sentiment detected by the sentiment detecting module **202** (negative in this case), the prompt generating module **211** provides an open-ended prompt **306**, e.g., "Today sounds like one of the days when you could use all the support. What happened, Alex?" to the user **102**. The user **102** may respond with **308** to the open-ended prompt **306**, e.g., "I'm sick. Down with covid". The emotion detecting AI model **206** detects emotion, e.g., 'unwell' of the user **102** and the medicalized term detecting AI model **208** detects a medicalized term, e.g., 'covid'. The conversational alignment monitoring module **212** of the AI model **112** increases the conversational alignment score for this conversation by 5 as the user **102** shared some context with the AI chatbot **110**.

[0074] The prompt generating module **211** provides an empathetic open-ended prompt **310** to the user **102** through the AI chatbot **110** based on the medicalized term, e.g., "I can imagine things are harder with covid. I understand how uncertainty can add on to the stress. Things may seem out of control but right now it's important to stay safe and aware. This too shall pass. Tell me more about this feeling". The user **102** may respond with **312** to the open-ended prompt

310 provided by the AI chatbot **110**, e.g., "I'm also a little lonely because my mother is scared of COVID." The emotion detecting AI model **206** detects the emotion, e.g., "lonely" of the user **102**. The conversational alignment monitoring module **212** increases the conversational alignment score by 5 to a total of 10 because the user shared further context.

[0075] The prompt generating module **211** generates and provides an empathetic closed-ended prompt **314** to the user **102** through the AI chatbot **110** based on the emotion of the user **102**, e.g., "Loneliness can feel like nobody understands us, or that we're disconnected from our own selves. And is this bringing up other feelings as well for you, Alex?" The user **102** may respond with **316** to the closed-ended prompt **314** provided by the AI chatbot **110**, e.g., "Yes, I feel ostracized." The conversational alignment monitoring module **212** of the AI model **112** increases the conversational alignment score by 5 to 15 because the user **102** shared more when prompted.

[0076] In FIG. **3**B of a user interface **301**, the prompt generating module **211** provides an open-ended prompt **318** to the user **102** through the AI chatbot **110** regarding negative feelings of the user **102**, e.g., "I'm sorry, that must be hard. What is a thought that comes to your mind when you feel this way? Your inner voice can sound like—'I'm boring', 'No one wants to talk to me', etc". The user **102** may respond with **320** to the open-ended prompt **318** provided by the AI chatbot **110**, e.g., "Like I'm an unwanted burden. It's my own fault". The conversational alignment monitoring module **212** updates the conversational alignment score for this user to 20.

[0077] The prompt generating module **211** generates and provides an empathetic closed-ended prompt **322** to the user **102** through the AI chatbot **110**, e.g., "Sometimes, it's helpful to remember that even though our thoughts may seem true, they may not be the reality. Would you agree?" The user **102** may respond with **324** to the closed-ended prompt **322** provided by the AI chatbot **110**, e.g., "I guess so." The conversational alignment monitoring module **212** of the AI model **112** increases the conversational alignment score to 25 because the user **102** agreed with the AI chatbot **110**. The conversational alignment score monitoring module **218** determines that the conversational alignment score, (e.g. 25) has reached a threshold of the conversational alignment score (25 in this case).

[0078] The alliance confirmation module **220** checks the user's readiness to use an intervention when the conversational alignment score exceeds the threshold by dynamically providing an alliance confirmation prompt **326** to the user **102**, e.g., "Thank you for sharing this with me. I may have something to help you manage your feelings around this. Would you like to give it a try?" The user **102** may accept the intervention by responding with **328** to the alliance confirmation prompt **326** provided by the AI chatbot **110**, e.g., "Yes, let's try."

[0079] FIGS. **4**A-**4**C are mock-up screenshots of user interfaces that illustrate a conversation between the AI chatbot **110** and the user **102** with one or more misalignments according to some embodiments herein. In FIG. **4**A of a user interface **400**, the AI chatbot **110** starts a conversation with the user **102** with a mood check **402**, e.g., "John, it's nice to see you! How's your day going so far?". The user **102** may respond with **404** to the AI chatbot **110**, e.g., "Not great." The sentiment detecting AI model **202** detects that the response **404** provided by the user **102** has a negative sentiment. Based on the sentiment detected, the AI chatbot **110** provides an open-ended prompt **406**, e.g., "What happened, John?" to the user **102**. The user **102** may respond with **408** to the open-ended prompt **406**, e.g., "I'm sad". The emotion detecting AI model **206** detects the emotion, e.g., "sad" of the user **102** from the response **408** received from the user **102**. The conversational alignment monitoring module **212** of the AI model **112** assigns the starting conversational alignment score as 5 because the user **102** shared some context.

[0080] The prompt generating module **211** provides an empathetic open-ended prompt **410** to the user **102** through the AI chatbot **110** based on the emotion, e.g., "Being sad is also okay. Sometimes things are not in our control. It will pass. What brought up these feelings for you?". The user **102** may respond with **412** to the open-ended prompt **410** provided by the AI chatbot **110**, e.g., "My family and my friends just don't get me." The domain detecting AI model **210** detects context, e.g., relationship of the user **102** in the response **412** received from the user **102**. The conversational alignment monitoring module **212** increases the conversational alignment score to 10 as the user **102** shares further context when prompted by the chatbot **110**.

[0081] The prompt generating module **211** provides an empathetic closed-ended prompt **414** to the user **102** through the AI chatbot **110** based on the detected domain, e.g., "It helps to talk about what's happening. You're going to be okay. Tell me more, John". The user **102** may respond with **416** to the closed-ended prompt **414** provided by the AI chatbot **110**, e.g., "You wouldn't understand me. You're just a bot." The misalignment detecting module **214** detects that there is misalignment in the conversation of the user **102** as the user **102** refused to share and showed a lack of trust in the AI chatbot **110**. Hence, the conversational alignment score is reduced to 5 for this major misalignment.

[0082] The conversation alignment score monitoring module **218** detects a drop in the conversational alignment between the user **102** and the AI chatbot **110** and attempts to recover from this misalignment and bring back the user **102** to the conversation by providing a recovery prompt **418** to the user **102** through the prompt generating module **211**, e.g., "I'm trying my best to understand you better. Try venting to me, it will help."

[0083] In FIG. **4**B of a user interface **401**, the user **102** may respond with **420** to the recovery prompt **418** provided by the AI chatbot **110**, e.g., "Ok, let's keep talking." The AI chatbot **110** recovered from the major misalignment as the user **102** agreed to continue with the conversation. Thus, the conversational alignment monitoring module **212** rewards this recovery by increasing the conversational alignment score to 12.

[0084] To bring the user **102** back to the conversation, the prompt generating module **211** generates an open-ended prompt **422** by replaying the response **412** shared by the user **102** before the misalignment was detected. The prompt generating module **211** provides the open-ended prompt **422** to the user **102** through the AI chatbot **110**, e.g., "Okay, you mentioned that your family and your friends just don't get you. Go on. I'm listening." The user **102** may respond with **424** to the open-ended prompt **422** provided by the AI chatbot **110**, e.g., "I really love them and I am trying to help but they always reject me and are angry at me." The domain detecting AI model **210** detects the context, e.g., relationship

of the user **102** in the response **424**. The conversational alignment monitoring module **212** increases the conversational alignment score for this user shared context to 17.

[0085] The prompt generating module **211** generates and provides an empathetic closed-ended prompt **426** regarding the detected domain to the user **102** through the AI chatbot **110**, e.g., "Hmm, relationships are complicated. Sometimes people close to us hurt us the most. And is this bringing up other feelings as well for you, John?" The user **102** may respond with **428** to the closed-ended prompt **426** provided by the AI chatbot **110**, e.g., "It makes me so angry." The conversational alignment monitoring module **212** increases the conversational alignment score for this user shared context to 22.

[0086] Further, the prompt generating module **211** generates and provides an empathetic closed-ended prompt **430** to the user **102** through the AI chatbot **110** based on the emotion detected in the response **428** of the user **102**, e.g., "It's natural to feel angry when you feel other people are making things worse rather than helping. Are there other issues like sleep, hunger or being exhausted that may be making you feel even worse?" The user **102** may respond with **432** to the closed-ended prompt **430** provided by the AI chatbot **110**, e.g., "I don't know." The misalignment detecting module **214** detects that there is a misalignment in the conversation of the user **102** as the user **102** expressed uncertainty about how to respond to the AI chatbot **110**. The conversational alignment monitoring module **212** reduces the conversational alignment score to 20 for this minor misalignment.

[0087] The prompt generating module **211** generates an open-ended prompt **434** based on the response **424** shared by the user **102** for the open-ended prompt **422**, i.e., by replaying the response **424** shared by the user **102** before the misalignment was detected. The prompt generating module **211** provides the open-ended prompt **434** to the user **102** through the AI chatbot **110**, e.g., "That's okay. Help me understand better, John. You mentioned that you really love them and are trying to help but they always reject you and are angry at you. What was the most difficult part to deal with?"

[0088] In FIG. **4**C of a user interface **403**, the user **102** may respond with **436** to the open-ended prompt **434** provided by the AI chatbot **110**, e.g., "That it's always me who has to compromise." The conversational alignment monitoring module **212** rewards this recovery from the minor misalignment by increasing the conversational alignment score to 24.

[0089] The prompt generating module **211** generates and provides an open-ended prompt **438** to the user **102** through the AI chatbot **110** based on the context detected earlier, i.e., the relationship domain detected in the response **424** of the user **102**, e.g., "Stressful situations can put pressure on most relationships. There is normally a thought or belief about ourselves or the other people involved in an event or situation that makes us feel the way we do. What was that for you?" The user **102** may respond with **440** expressing confusion to the open-ended prompt **438** provided by the AI chatbot **110**, e.g., "What do you mean?" The misalignment detecting module **214** detects that there is a misalignment in the conversation of the user **102** as the user **102** didn't understand what the AI chatbot **110** was saying or asking. The conversational alignment monitoring module **212** reduces the conversational alignment score to 22 for this minor misalignment.

[0090] The AI model **112** recovers from this misalignment and brings back the user **102** to the conversation by providing an explanation and a recovery prompt **442** to the user **102** through the AI chatbot **110**, e.g., "Some situations cause us to react with a feeling or a behavior which is often triggered by an automatic thought in our mind. For example, meeting new people may make us think that they won't like us which, in turn, might trigger sad or anxious feelings in us. Was there such a thought for you?"

[0091] The user **102** may respond with **444** to the recovery prompt **442** provided by the AI chatbot **110**, e.g., "Yes, I think that maybe they don't love me." Since the AI chatbot **110** succeeded in getting the user **102** to share more, the conversational alignment monitoring module **212** rewards this recovery from a minor misalignment by increasing the conversational alignment score to 26.

[0092] The conversational alignment score monitoring module **218** detects that the conversational alignment score has exceeded the threshold (25 in this case). Thus, the alliance confirmation module **220** provides an alliance confirmation prompt **446** to the user **102** through the AI chatbot **110** based on the response **444** of the user **102**, e.g., "Thank you for sharing this with me. Thoughts like that can be hard to shake. I may have something to help you manage your feelings around this. Would you like to give it a try?" The user **102** may accept the intervention by responding with **448** to the alliance confirmation prompt **446** provided by the AI chatbot **110**, e.g., "Yes, let's try."

[0093] FIG. **4**D is an exemplary graphical representation **405** of the conversational alignment score according to some embodiments herein. In the exemplary graphical representation **405**, indices of messages exchanged between the user **102** and the AI chatbot **110** are plotted in a X-axis and conversational alignment scores are plotted in a Y-axis. The exemplary graphical representation **405** depicts the conversation between the user **102** and the AI chatbot **110** includes one or more misalignments when the user **102** (*i*) expressed lack of trust in the AI chatbot **110**, (ii) expressed uncertainty about how to respond to the AI chatbot **110**, and (iii) didn't understand what the AI chatbot **110** was saying or asking.

[0094] In some embodiments, the conversational alignment score is increased by the conversational alignment monitoring module **212** if at least one of the emotion is detected by the emotion detecting AI model **206**, (ii) medicalized terms are detected using the medicalized term detecting AI model **208**, and (iii) domain is detected using the domain detecting AI model **210**. In some embodiments, the conversational alignment score is increased by the conversational alignment monitoring module **212** if the alignment detecting module **214** detects that the user **102** is agreeing with the AI chatbot **110**. In some embodiments, the conversational alignment score is reduced by the conversational alignment monitoring module **212** if the misalignment detecting module **216** detects that the user **102** (*i*) expresses lack of trust in the AI chatbot **110**, (ii) expresses uncertainty about how to respond to the AI chatbot **110**, (iii) didn't understand what the AI chatbot **110** was saying or asking, (iv) disagrees with the AI chatbot **110**, and (v) is unhappy or dissatisfied with the AI chatbot **110**. In some embodiments, the conversational alignment score is updated after each response received from the user **102** at the AI chatbot **110**

during the conversation and indicates strength of the conversational alignment formed between the AI chatbot **110** and the user **102**.

[0095] FIGS. **5**A-**5**C are mock-up screenshots of user interfaces that illustrate a conversation between the user and the AI chatbot according to some embodiments herein. In FIG. **5**A of a user interface **500**, the AI chatbot **110** starts a conversation with the user **102** with a mood check **502**, e.g., "Hello, welcome! Thank you for visiting our website. How are you doing today?" The user **102** may respond with **504** to the open-ended prompt **502** provided by the AI chatbot **110**, e.g., "I'm good, thanks for asking." The prompt generating module **211** provides an open-ended prompt **506** based on the sentiment detected in the user response **504** by the sentiment detection module **202** (positive in this case), e.g., "How can we assist you?" to the user **102**. The user **102** may respond with **508** to the open-ended prompt **506**, e.g., "I would like to return a shirt I recently bought from you.". The conversational alignment monitoring module **212** detects that the user **102** shared some context and assigns the starting conversational alignment score as 5.

[0096] The prompt generating module **211** provides an open-ended prompt **510**, e.g., "We need some additional information prior to proceeding. Can you please share the order number?" to the user **102** based on the response **508**. The user **102** may respond with **512** to the open-ended prompt **510**, e.g., "964823FPX". The conversational alignment monitoring module detects that the user **102** went with the conversational flow and responded to the open-ended prompt **510** as expected. The conversational alignment monitoring module **212** increases the conversational alignment score for this user shared context to 10.

[0097] The prompt generating module **211** provides an open-ended prompt **514**, e.g., "Thank you for that. Can you describe the issue you're facing with the shirt?" to the user **102** based on the response **512**. The user **102** may respond with **516** to the open-ended prompt **514**, e.g., "The cloth quality is not up to the mark". The conversational alignment monitoring module detects that the user **102** shared more context when prompted. The conversational alignment monitoring module **212** increases the conversational alignment score to 15.

[0098] The prompt generating module **211** provides an empathetic open-ended prompt **518**, e.g., "We apologize for the inconvenience you have faced. Do you want me to mark it for return?", to the user **102** based on the response **516**. The user **102** may respond with **520** to the open-ended prompt **518**, e.g., "Yes, please.". The conversational alignment monitoring module **212** detects that the user **102** agreed with the AI chatbot **110**. The conversational alignment monitoring module **212** of the AI model **112** increases the conversational alignment score to 20.

[0099] The prompt generating module **211** provides an open-ended prompt **522**, e.g., "The amount will be credited to your account within 5 business days after the item is picked up. Can you confirm the address for pickup?" to the user **102** based on the response **520**. The user **102** may respond with **524** to the open-ended prompt **522**, e.g., "Can't you give me a refund instead?". The misalignment detecting module **216** detects that there is a minor misalignment in the conversation between the user **102** and the AI chatbot **110** as the user **12** didn't share the information they were requested for with the AI chatbot **110**. The conversational alignment monitoring module **212** reduces the conversational alignment score to 18.

[0100] In FIG. **5**B of a user interface **501**, the prompt generating module **211** provides a recovery prompt **526**, e.g., "We sincerely apologize, but we don't have a refund option available at present. The amount can be credited to your account and used to buy something else." to the user **102** based on the response **524**. The user **102** may respond with **528** to the recovery prompt **526**, e.g., "That's not acceptable at all. I will have to unnecessarily purchase something because of a low-quality product sent by you!" The misalignment detecting module **216** detects that there is a major misalignment in the conversation between the user **102** and the AI chatbot **110** as the user **102** is unhappy with the AI chatbot **110**. The conversational alignment monitoring module **212** reduces the conversational alignment score to 13.

[0101] The prompt generating module **211** provides an empathetic recovery prompt **530**, e.g., "We're very sorry. We will take care of this on top priority for you. Let me have a word with my manager to find out how we can fix this for you. Can I put you on hold for some time?" to the user **102** based on the response **528** for the recovery prompt **526**. The user **102** may respond **532** to the recovery prompt **530**, e.g., "Okay". The alignment detecting module **214** detects that the user **102** responded positively to the AI chatbot **110** and wants to continue the conversation. The conversational alignment monitoring module **212** rewards this recovery by increasing the conversational alignment score to 23.

[0102] The prompt generating module **211** provides a closed-ended prompt **534**, e.g., "Thanks for your patience. Unfortunately, we can't offer you a refund but we can give you a coupon code worth a discount of 20% on any future purchase. Would you like to do that?" to the user **102** based on the response **532**. The user **102** may respond with **536** to the open-ended prompt **534**, e.g., "I'm not sure.". The misalignment detecting module **216** detects that there is a minor misalignment in the conversation between the user **102** and the AI chatbot **110** as the user **102** expressed uncertainty about how to respond to the AI chatbot **110**. The conversational alignment monitoring module **212** reduces the conversational alignment score to 21.

[0103] The prompt generating module **211** provides a closed-ended prompt **538**, e.g., "I understand. I would like to tell you that the coupon has unlimited validity and can be used on any item. However, it cannot be combined with other offers. Do you need any more details about this coupon?" to the user **102** based on the response **536** for the open-ended prompt **534**. The user **102** may respond with **540** to the closed-ended prompt **538**, e.g., "No, I don't think so". The alignment detecting module **214** detects that the conversation between the user **102** and the AI chatbot **110** is aligned. The conversational alignment monitoring module **212** rewards the recovery by increasing the conversational alignment score to 25.

[0104] In FIG. **5**C of a user interface **503**, the conversational alignment score monitoring module **212** detects that the conversational alignment score has reached the threshold (25 in this case). Hence, the alliance confirmation module **220** provides an alliance confirmation prompt **542**, e.g., "So would you like to accept this coupon?" to the user **102** based on the response **540**. The user **102** may respond with **544** to the closed-ended prompt **542**, e.g., "Yes, alright". The alignment detecting module **214** detects that the user **102**

accepted the solution suggested by the AI chatbot **110**. The AI chatbot **110** rounds up the conversation by asking the user **102** for their return pickup address through **546** and completing the transaction through **550** and **554**. In some embodiments, FIGS. **5A-5C** illustrate a different chatbot application which is more transactional in nature and may not require giving the user **102** a space to vent. In some embodiments, a threshold can be designed according to the chatbot application.

[0105] FIG. **6** is a flowchart that illustrates a method **600** for monitoring a conversational alignment and establishing an alliance between the artificial intelligence (AI) chatbot **110** and the user **102** according to some embodiments herein. At a step **602**, a conversation is started with the AI chatbot **110** by the user **102**. At a step **604**, sentiment of the user's opening text is detected and an empathetic statement is provided to the user **102** based on the sentiment by the AI model **112**. At a step **606**, one or more contextual features are extracted by the AI model **112**. In some embodiments, the one or more contextual features are stored in the database **200** of the server **108**. At the step **614**, it is checked if the extracted features indicate that the user **102** is in crisis. If YES, control goes to step **624** or else goes to step **616**. At the step **616**, it is checked if the user **102** responded in alignment with what the AI chatbot **110** said or asked. If Yes, control goes to step **618** or the conversational alignment score is reduced at step **626** and the user's concern is addressed, empathy is shown, and brings the user **102** back to the conversation at step **628**. At the step **618**, the conversational alignment score is increased if the user **102** responded in alignment with what the AI chatbot **110** said or asked. At the step **620**, it is checked if the conversational alignment score exceeds a threshold, if YES, control goes to step **622** or else goes to step **630**. At step **622**, an alliance confirmation prompt is provided to the user **102** by the server **108** through the AI chatbot **110** to confirm alliance and user's readiness to accept an intervention. At a step **624**, an intervention is started. At a step **630**, an empathetic response, an open-ended or closed-ended prompt based on context gathered.

[0106] FIG. **7** is a flow diagram that illustrates a method **700** for monitoring and improving conversational alignment to develop an alliance between the artificial intelligence (AI) chatbot **110** and the user **102** according to some embodiments herein. At a step **702**, the method **700** includes providing a first prompt to the user **102** by the AI chatbot **110** to obtain a first response from the user **102** for the first prompt. The first response is at least one of a text input or a voice input. At a step **704**, the method **700** includes extracting sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response. Extracting the at least one contextual feature includes detecting that the first response of the user **102** includes at least one of emotion, medicalized terms or domain. The emotion is detected using the emotion detecting artificial intelligence (AI) model **206**, the medicalized terms are detected using the medicalized term detecting AI model **208**, and the domain is detected using the domain detecting AI model **210**.

[0107] At a step **706**, the method **700** includes generating a second prompt based on the at least one contextual feature that includes at least one of the emotion, the medicalized terms or the domain. At a step **708**, the method **700** includes determining, using the AI model **112**, if a conversation between the user **102** and the AI chatbot **110** has a conversational alignment. The conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot **110** and agreeing to suggestions or interpretations made by the AI chatbot **110**. At a step **710**, the method **700** includes increasing a conversational alignment score for a second response of the user **102** in the conversation if, using the AI model **112**, the conversational alignment is determined. At a step **712**, the method **700** includes monitoring, using the AI model **112**, the conversation to determine if there is a misalignment in the conversation between the user **102** and the AI chatbot **110** and reduce the conversational alignment score if the misalignment is detected. At a step **714**, the method **700** includes determining, using the AI model **112**, a type of misalignment and generating a recovery prompt to recover from the misalignment. At a step **716**, the method **700** includes increasing the conversational alignment score for a third response from the user **102** for the recovery prompt if the conversation is recovered from the misalignment and, using the AI model **112**, the conversational alignment is determined for the third response. At a step **718**, the method **700** includes determining that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold. At a step **720**, the method **700** includes generating an alliance confirmation prompt to confirm establishment of the alliance with the user **102** when the conversational alignment score reaches the threshold.

[0108] The embodiments herein may include a computer program product configured to include a pre-configured set of instructions, which when performed, can result in actions as stated in conjunction with the methods described above. In an example, the pre-configured set of instructions can be stored on a tangible non-transitory computer readable medium or a program storage device. In an example, the tangible non-transitory computer readable medium can be configured to include the set of instructions, which when performed by a device, can cause the device to perform acts similar to the ones described here. Embodiments herein may also include tangible and/or non-transitory computer-readable storage media for carrying or having computer executable instructions or data structures stored thereon.

[0109] Generally, program modules utilized herein include routines, programs, components, data structures, objects, and the functions inherent in the design of special-purpose processors, etc. that perform particular tasks or implement particular abstract data types. Computer executable instructions, associated data structures, and program modules represent examples of the program code means for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps. The embodiments herein can include both hardware and software elements. The embodiments that are implemented in software include but are not limited to, firmware, resident software, microcode, etc. A data processing system suitable for storing and/or executing program code will include at least one processor coupled directly or indirectly to memory elements through a system bus. The memory elements can include local memory employed during actual execution of the program code, bulk storage, and cache memories which provide temporary storage of at least some program code in order to reduce the number of times code must be retrieved from bulk storage during execution.

[0110] Input/output (I/O) devices (including but not limited to keyboards, displays, pointing devices, etc.) can be coupled to the system either directly or through intervening I/O controllers. Network adapters may also be coupled to the system to enable the data processing system to become coupled to other data processing systems or remote printers or storage devices through intervening private or public networks. Modems, cable modem and Ethernet cards are just a few of the currently available types of network adapters.

[0111] A representative hardware environment for practicing the embodiments herein is depicted in FIG. **8**, with reference to FIGS. **1** through **7**. This schematic drawing illustrates a hardware configuration of the AI model **112**/a computer system/a user device **104** in accordance with the embodiments herein. The user device **104** includes at least one processing device **10** and a cryptographic processor **11**. The special-purpose CPU **10** and the cryptographic processor (CP) **11** may be interconnected via system bus **14** to various devices such as a random access memory (RAM) **15**, read-only memory (ROM) **16**, and an input/output (I/O) adapter **17**. The I/O adapter **17** can connect to peripheral devices, such as disk units **12** and tape drives **13**, or other program storage devices that are readable by the system. The user device **104** can read the inventive instructions on the program storage devices and follow these instructions to execute the methodology of the embodiments herein. The user device **104** further includes a user interface adapter **20** that connects a keyboard **18**, mouse **19**, speaker **25**, microphone **23**, and/or other user interface devices such as a touch screen device (not shown) to the bus **14** to gather user input. Additionally, a communication adapter **21** connects the bus **14** to a data processing network **26**, and a display adapter **22** connects the bus **14** to a display device **24**, which provides a graphical user interface (GUI) **30** of the output data in accordance with the embodiments herein, or which may be embodied as an output device such as a monitor, printer, or transmitter, for example. Further, a transceiver **27**, a signal comparator **28**, and a signal converter **29** may be connected with the bus **14** for processing, transmission, receipt, comparison, and conversion of electric or electronic signals.

[0112] The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others can, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein can be practiced with modification within the spirit and scope of the appended claims.

1. A processor-implemented method for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user, comprising:

dynamically generating, using an artificial intelligence (AI) model at the server, a first prompt based on context gathered from a user device associated with the user;

automatically providing, using the AI model at the server, the first prompt to the user by the AI chatbot to obtain a first response from the user device associated with the user for the first prompt, wherein the first response is at least one of text or voice;

training the AI model at the server by vectorizing the text received from the user to convert the text into a numerical representation to use the numerical representation corresponding to the text;

extracting, using the AI model at the server, sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response by detecting that the first response of the user comprises at least one of emotion using an emotion detecting artificial intelligence (AI) model, medicalized terms using a medicalized term detecting AI model or domain using a domain detecting AI model;

generating, using the AI model at the server, a second prompt based on the at least one contextual feature that comprises at least one of the emotion, the medicalized terms, or the domain;

determining, using the AI model at the server, if a conversation between the user and the AI chatbot has a conversational alignment to increase a conversational alignment score for a second response of the user in the conversation if the conversational alignment is determined, wherein the conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot and agreeing to suggestions or interpretations made by the AI chatbot;

continuously monitoring, using the AI model at the server, the conversation to determine if there is a misalignment in the conversation between the user and the AI chatbot and reduce the conversational alignment score if the misalignment is detected;

training a plurality of machine learning models based on training data that comprises user text that is labeled as true, and the user text labeled as false, an intent for each type of misalignment for intent recognition by providing representative patterns for each type of misalignment;

classifying, using the plurality of machine learning models at the server, a type of misalignment from a plurality of misalignments;

generating a recovery prompt to recover from the misalignment based on the type of misalignment that is classified by the plurality of machine learning models at the server;

increasing, using the AI model at the server, the conversational alignment score for a third response from the user for the recovery prompt if the AI model at the server determines that there is conversational alignment for the third response;

determining, using the AI model at the server, that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold; and

generating, using the AI model at the server, an alliance confirmation prompt to confirm establishment of the alliance with the user when the conversational alignment score reaches the threshold.

2. The method of claim **1**, further comprising if a plurality of contextual features are detected in each response of the user, prioritizing the plurality of contextual features to decide which direction to take the conversation, wherein the

plurality of contextual features are prioritized as (i) the at least one medicalized term, (ii) domain, and (iii) emotion.

3. The method of claim **1**, wherein prompts are generated based on a contextual feature that has highest priority among the plurality of contextual features, wherein the contextual feature that has the highest priority is identified by prioritizing the plurality of contextual features in a decreasing order.

4. The method of claim **1**, wherein the type of misalignment is selected from at least one of confusion, disagreement, dissatisfaction, lack of trust, refusal or uncertainty expressed by the user to the AI chatbot.

5. The method of claim **4**, wherein at least one of the confusion, the disagreement, the dissatisfaction, the lack of trust, the refusal or uncertainty expressed by the user to the AI chatbot is detected using the intent recognition AI model.

6. (canceled)

7. (canceled)

8. The method of claim **5**, wherein if a confidence score for the matching pattern with a highest confidence score is above an intent matching threshold, a response received from the user is determined to correspond to the intent that the matching pattern represents.

9. (canceled)

10. The method of claim **1**, wherein the conversational alignment score is updated after each response received from the user at the AI chatbot during the conversation and indicates strength of the conversational alignment formed between the AI chatbot and the user.

11. The method of claim **1**, wherein the text input is vectorized using frequency-based techniques or semantics-based techniques.

12. The method of claim **1**, wherein the prompts are generated based on predefined base prompts that are written by conversation designers and parameterized with the user's context to personalize them, wherein the predefined base prompts are stored in a database of a server.

13. The method of claim **1**, wherein the prompts are provided to the user through the AI chatbot until the conversational alignment score exceeds the threshold.

14. The method of claim **1**, wherein at least one of the open-ended prompts or the closed-ended prompts are alternately provided to the user based on the conversational alignment score.

15. The method of claim **1**, further comprising recommending at least one intervention to the user when the alliance has been confirmed using the alliance confirmation prompt in which the user agrees to try out an intervention.

16. One or more non-transitory computer readable storage mediums storing one or more sequences of instructions, which when executed by one or more processors, causes a method for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user by performing the steps of:

dynamically generating, using an artificial intelligence (AI) model at the server, a first prompt based on context gathered from a user device associated with the user;

automatically providing, using the AI model at the server, the first prompt to the user by the AI chatbot to obtain a first response from the user device associated with the user for the first prompt, wherein the first response is at least one of text or voice;

training the AI model at the server by vectorizing the text received from the user to convert the text into a numerical representation to use the numerical representation corresponding to the text;

extracting, using the AI model at the server, sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response-by detecting that the first response of the user comprises at least one of emotion using an emotion detecting artificial intelligence (AI) model, medicalized terms using a medicalized term detecting AI model or domain using a domain detecting AI model;

generating, using the AI model at the server, a second prompt based on the at least one contextual feature that comprises at least one of the emotion, the medicalized terms, or the domain;

determining, using the AI model at the server, if a conversation between the user and the AI chatbot has a conversational alignment to increase a conversational alignment score for a second response of the user in the conversation if the conversational alignment is determined, wherein the conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot and agreeing to suggestions or interpretations made by the AI chatbot;

continuously monitoring, using the AI model at the server, the conversation to determine if there is a misalignment in the conversation between the user and the AI chatbot and reduce the conversational alignment score if the misalignment is detected;

training a plurality of machine learning models based on training data that comprises user text that is labeled as true, and the user text labeled as false, an intent for each type of misalignment for intent recognition by providing representative patterns for each type of misalignment;

classifying, using the plurality of machine learning models at the server, a type of misalignment from a plurality of misalignments;

generating a recovery prompt to recover from the misalignment based on the type of misalignment that is classified by the plurality of machine learning models at the server;

increasing, using the AI model at the server, the conversational alignment score for a third response from the user for the recovery prompt if the AI model at the server determines that there is conversational alignment for the third response;

determining, using the AI model at the server, that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold; and

generating, using the AI model at the server, an alliance confirmation prompt to confirm establishment of the alliance with the user when the conversational alignment score reaches the threshold.

17. The one or more non-transitory computer readable storage mediums storing the one or more sequences of instructions of claim **16**, further comprising if a plurality of contextual features are detected in each response of the user, prioritizing the plurality of contextual features to decide which direction to take the conversation, wherein the plurality of contextual features are prioritized as (i) the at least one medicalized term, (ii) domain, and (iii) emotion.

18. The one or more non-transitory computer readable storage mediums storing the one or more sequences of

instructions of claim **16**, wherein prompts are generated based on a contextual feature that has highest priority among the plurality of contextual features, wherein the contextual feature that has the highest priority is identified by prioritizing the plurality of contextual features in a decreasing order.

19. (canceled)

20. A system for monitoring and improving conversational alignment to develop an alliance between an artificial intelligence (AI) chatbot and a user comprising:

a device processor; and

a non-transitory computer readable storage medium storing one or more sequences of instructions, which when executed by the device processor, causes a method by performing the steps of:

dynamically generating, using an artificial intelligence (AI) model at the server, a first prompt based on context gathered from a user device associated with the user;

automatically providing, using the AI model at the server, the first prompt to the user by the AI chatbot to obtain a first response from the user device associated with the user for the first prompt, wherein the first response is at least one of text or voice;

training the AI model at the server by vectorizing the text received from the user to convert the text into a numerical representation to use the numerical representation corresponding to the text;

extracting, using the AI model at the server, sentiment or at least one contextual feature from the first response when the at least one contextual feature is present in the first response-by detecting that the first response of the user comprises at least one of emotion using an emotion detecting artificial intelligence (AI) model, medicalized terms using a medicalized term detecting AI model or domain using a domain detecting AI model;

generating, using the AI model at the server, a second prompt based on the at least one contextual feature that comprises at least one of the emotion, the medicalized terms, or the domain;

determining, using the AI model at the server, if a conversation between the user and the AI chatbot has a conversational alignment to increase a conversational alignment score for a second response of the user in the conversation if the conversational alignment is determined, wherein the conversational alignment is an alignment with respect to the user sharing more context with the AI chatbot and agreeing to suggestions or interpretations made by the AI chatbot;

continuously monitoring, using the AI model at the server, the conversation to determine if there is a misalignment in the conversation between the user and the AI chatbot and reduce the conversational alignment score if the misalignment is detected;

training a plurality of machine learning models based on training data that comprises user text that is labeled as true, and the user text labeled as false, an intent for each type of misalignment for intent recognition by providing representative patterns for each type of misalignment;

classifying, using the plurality of machine learning models at the server, a type of misalignment from a plurality of misalignments;

generating a recovery prompt to recover from the misalignment based on the type of misalignment that is classified by the plurality of machine learning models at the server;

increasing, using the AI model at the server, the conversational alignment score for a third response from the user for the recovery prompt if the AI model at the server determines that there is conversational alignment for the third response;

determining, using the AI model at the server, that the conversational alignment score exceeds a threshold by comparing the conversational alignment score with the threshold; and

generating, using the AI model at the server, an alliance confirmation prompt to confirm establishment of the alliance with the user when the conversational alignment score reaches the threshold.

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